

CLASSICAL & RIGID BODY MECHANICS

Complete Master Lecture Notes & Solved Question Manual

Official Publication of sciqb.com

Typeset into Complete LaTeX Notes by Aryan (sciqb.com)

July 24, 2026

Contents

1	Laws of Vectors in 2D & System of Coplanar Forces	3
1.1	Vectors in 2D and Vector Algebra in Mechanics	3
1.2	Coplanar Concurrent Forces	4
1.3	System of Parallel Forces and Center of Parallel Forces	6
1.4	Theory of Couples	6
1.5	Moments of a Force and Varignon's Theorem	8
1.6	Reduction of General Coplanar System of Forces	9
1.7	Laws of Friction and Inclined Planes	10
1.8	Solved Problems	10
2	Principle of Virtual Work (Statics)	13
2.1	Concept of Virtual Displacement and Virtual Work	13
2.2	Principle of Virtual Work	13
2.2.1	Active Forces vs. Forces Which Do No Work	13
2.3	Hooke's Law for Elastic Strings	14
2.4	Stability of Equilibrium	14
2.5	Worked Examples (Complete Unabridged Collection)	15
3	General Motion in Two Dimensions (Kinematics)	19
3.1	Cartesian Coordinate System (x, y)	19
3.2	Polar Coordinate System (r, θ)	20
3.3	Intrinsic Coordinate System (s, ψ)	22
3.4	Worked Examples (Complete Unabridged Collection)	24
4	Simple Harmonic Motion (Dynamics)	27
4.1	Introduction and Governing Differential Equation	27
4.2	Kinematic Expressions	27
4.2.1	Velocity as a Function of Position	27
4.2.2	Displacement as a Function of Time	28
4.3	Important Properties of SHM	28
4.4	Worked Examples (Complete Unabridged Collection)	29
5	Central Orbits and Kepler's Laws (Dynamics)	35
5.1	Central Forces and Motion in a Plane	35
5.2	Fundamental Equations of Motion	35
5.3	Differential Equations of Central Orbits	35
5.3.1	Polar Form $(u = 1/r)$	35
5.3.2	Pedal Form $(p = r \sin \phi)$	36
5.4	Apses and Apisidal Distances	36
5.5	Inverse Square Law Motion $(F = \mu/r^2)$	36
5.6	Kepler's Laws of Planetary Motion	37
5.7	Worked Examples (Complete Unabridged Collection)	37

6	Constrained Motion on Vertical Curves	39
6.1	Motion on a Smooth Vertical Circle	39
6.2	Motion on a Smooth Vertical Cycloid	40
7	Motion in a Moving Frame of Reference (Dynamics)	41
7.1	Inertial vs. Non-Inertial Frames	41
7.2	Motion in a Linearly Accelerating Frame	41
7.3	Motion in a Uniformly Rotating Frame	41
7.3.1	Transformation of Velocity	41
7.3.2	Transformation of Acceleration	42
7.4	Equations of Motion and Pseudo Forces	42
7.5	Worked Examples	43
8	Forces in 3D, Wrench & Moment of Inertia	44
8.1	Reduction of 3D Systems of Forces	44
8.2	Dyname, Wrench and Poinsot's Central Axis	44
8.3	Moment of Inertia and Product of Inertia	45
8.4	Moment of Inertia and Principal Axes	46
8.5	Standard Moments of Inertia Table	46
9	Master Formula Cheat Sheet	47
9.1	Statics, Friction & Virtual Work	47
9.2	Common Catenary & Kinematics	48
9.3	Simple Harmonic Motion & Central Orbits	48
9.4	Constrained Motion, 3D Rigid Bodies & Moving Frames	49

Chapter 1

Laws of Vectors in 2D & System of Coplanar Forces

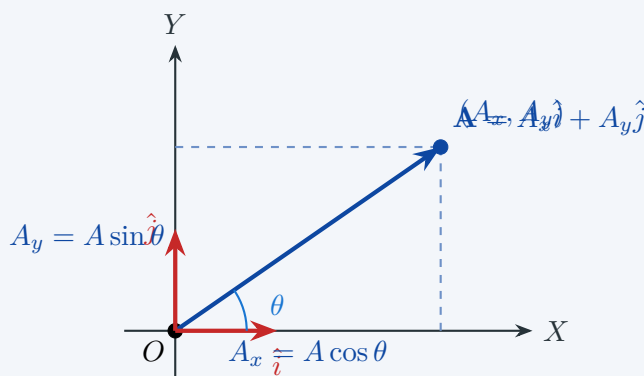
1.1 Vectors in 2D and Vector Algebra in Mechanics

Definition: Vector Representation in 2D

A physical quantity having both magnitude and direction is a **vector**. In a two-dimensional rectangular Cartesian coordinate system (X, Y) with unit vectors $\hat{i}$ and $\hat{j}$ along coordinate axes:

$$\mathbf{A} = A_x \hat{i} + A_y \hat{j} \quad (1.1.1)$$

where $A_x = A \cos \theta$ and $A_y = A \sin \theta$. Magnitude $|\mathbf{A}| = \sqrt{A_x^2 + A_y^2}$ and direction angle $\theta = \tan^{-1}(A_y/A_x)$.



Theorem / Fundamental Law: Vector Operations in Mechanics

1. Scalar (Dot) Product:

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y = AB \cos \theta \quad (1.1.2)$$

Work done by force $\mathbf{F}$ along displacement $\delta \mathbf{r}$ is $W = \mathbf{F} \cdot \delta \mathbf{r}$.

2. Vector (Cross) Product:

$$\mathbf{A} \times \mathbf{B} = (A_x B_y - A_y B_x) \hat{k} = AB \sin \theta \hat{k} \quad (1.1.3)$$

The moment of force $\mathbf{F}$ acting at position $\mathbf{r}$ relative to origin O is $\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$.

1.2 Coplanar Concurrent Forces

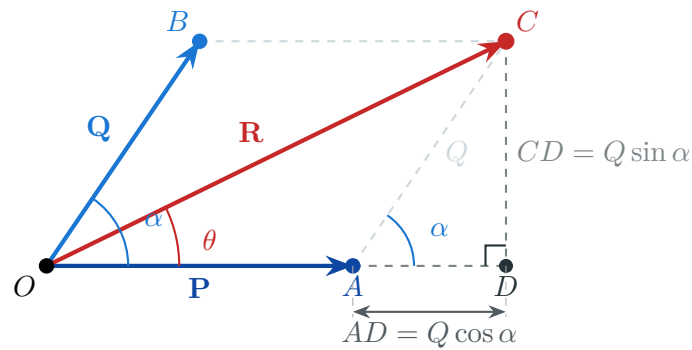
Theorem / Fundamental Law: Parallelogram Law of Forces

If two coplanar forces **P** and **Q** acting simultaneously on a particle at point *O* at angle α are represented in magnitude and direction by two adjacent sides of a parallelogram, their resultant **R** is represented in magnitude and direction by the diagonal passing through *O*:

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}, \quad \tan \theta = \frac{Q \sin \alpha}{P + Q \cos \alpha} \quad (1.2.1)$$

Rigorous Derivation / Proof: Proof of Parallelogram Law

Let *OA* represent force *P* along *X*-axis and *OB* represent force *Q* at angle α to *OA*. Complete parallelogram *OACB*. Draw perpendicular *CD* onto *OA* produced.



In right-angled triangle $\triangle ADC$:

$$AD = Q \cos \alpha, \quad CD = Q \sin \alpha$$

In right-angled triangle $\triangle ODC$:

$$OC^2 = OD^2 + CD^2 = (OA + AD)^2 + CD^2 = (P + Q \cos \alpha)^2 + (Q \sin \alpha)^2$$

$$R^2 = P^2 + 2PQ \cos \alpha + Q^2(\cos^2 \alpha + \sin^2 \alpha) = P^2 + Q^2 + 2PQ \cos \alpha$$

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}, \quad \tan \theta = \frac{CD}{OD} = \frac{Q \sin \alpha}{P + Q \cos \alpha}$$

Theorem / Fundamental Law: Lami's Theorem

If three coplanar concurrent forces **P**, **Q**, **R** acting at a point *O* are in static equilibrium, each force is proportional to the sine of the angle between the other two forces:

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma} \quad (1.2.2)$$

where $\alpha = \angle(\mathbf{Q}, \mathbf{R})$, $\beta = \angle(\mathbf{P}, \mathbf{R})$, and $\gamma = \angle(\mathbf{P}, \mathbf{Q})$.

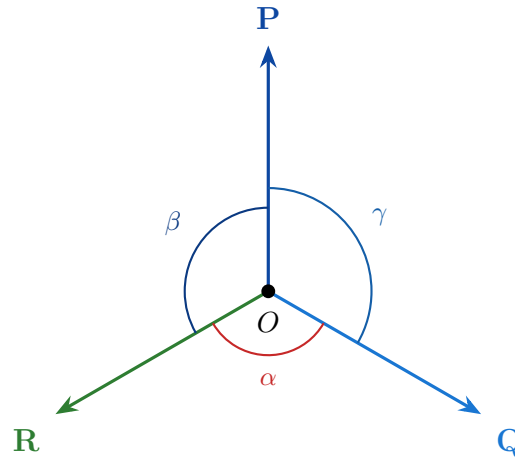


Figure 1.1: Lami's Theorem force vector diagram.

Rigorous Derivation / Proof: Proof of Lami's Theorem

Since forces $\mathbf{P}, \mathbf{Q}, \mathbf{R}$ are in equilibrium:

$$\mathbf{P} + \mathbf{Q} + \mathbf{R} = \mathbf{0} \implies \mathbf{P} + \mathbf{Q} = -\mathbf{R}$$

Taking vector cross product with $\mathbf{P}$:

$$\mathbf{P} \times (\mathbf{P} + \mathbf{Q} + \mathbf{R}) = \mathbf{0} \implies \mathbf{P} \times \mathbf{Q} = \mathbf{R} \times \mathbf{P}$$

Similarly, taking vector cross product with $\mathbf{Q}$:

$$\mathbf{Q} \times \mathbf{R} = \mathbf{P} \times \mathbf{Q}$$

Evaluating magnitudes:

$$PQ \sin(180^\circ - \gamma) = QR \sin(180^\circ - \alpha) = RP \sin(180^\circ - \beta)$$

Dividing by PQR :

$$\frac{\sin \gamma}{R} = \frac{\sin \alpha}{P} = \frac{\sin \beta}{Q} \implies \frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$$

1.3 System of Parallel Forces and Center of Parallel Forces

Theorem / Fundamental Law: Resultant and Center of Parallel Forces

1. **Like Parallel Forces** P and Q : Resultant $R = P + Q$ acting parallel to P and Q at point C dividing line segment AB internally in ratio:

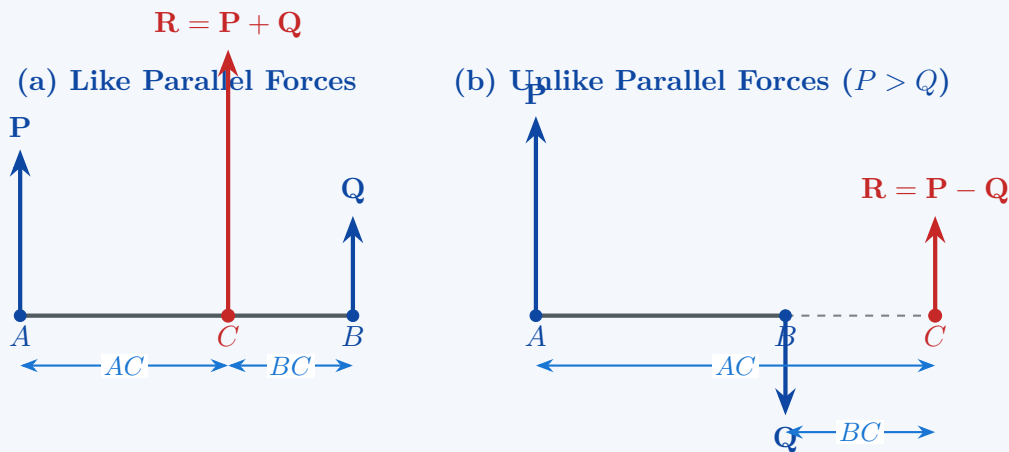
$$P \cdot AC = Q \cdot BC \implies \frac{AC}{BC} = \frac{Q}{P} \quad (1.3.1)$$

2. **Unlike Parallel Forces** $P > Q$: Resultant $R = P - Q$ acting parallel to P at point C dividing line segment AB externally in ratio:

$$P \cdot AC = Q \cdot BC \implies \frac{AC}{BC} = \frac{Q}{P} \quad (1.3.2)$$

3. **Center of Parallel Forces**: Position vector $\mathbf{r}_C$ of center of parallel forces $\mathbf{F}_i$ acting at points $\mathbf{r}_i$:

$$\mathbf{r}_C = \frac{\sum_{i=1}^n F_i \mathbf{r}_i}{\sum_{i=1}^n F_i} \quad (1.3.3)$$



1.4 Theory of Couples

Definition: Couple and Arm of Couple

A **couple** consists of two equal, opposite, and non-collinear parallel forces $(P, -P)$. - Vector sum of forces is zero $(\mathbf{P} + (-\mathbf{P}) = \mathbf{0})$. A couple produces pure rotation. - **Arm of Couple** (d): Perpendicular distance between lines of action of the two forces. - **Moment of Couple**: $G = P \cdot d$.

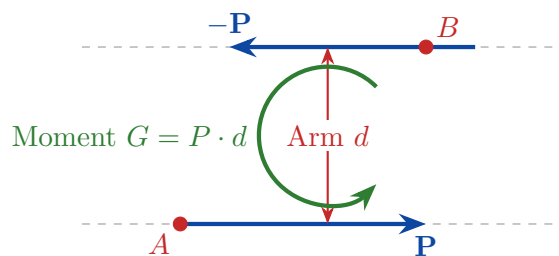
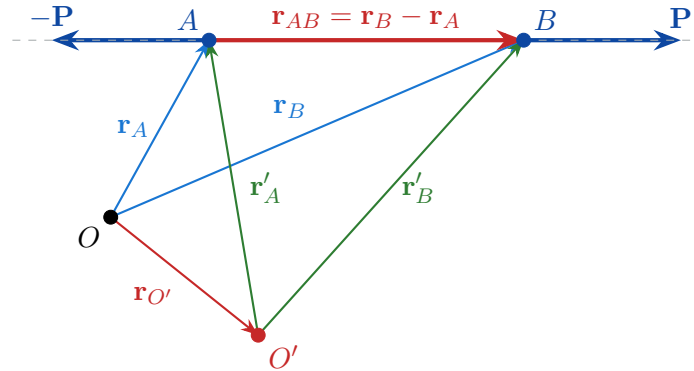


Figure 1.2: Vector representation of a couple $(P, -P)$ and its arm d , producing pure moment $G = P \cdot d$.

Rigorous Derivation / Proof: Proof of Couple Moment Invariance (Origin-Independence of Couple)

Let a couple consist of force $-\mathbf{P}$ acting at point $A(\mathbf{r}_A)$ and force $+\mathbf{P}$ acting at point $B(\mathbf{r}_B)$ relative to an arbitrary origin O .



1. Moment About Origin O :

The total vector moment of the couple about origin O is:

$$\mathbf{G}_O = \mathbf{r}_A \times (-\mathbf{P}) + \mathbf{r}_B \times \mathbf{P} = (\mathbf{r}_B - \mathbf{r}_A) \times \mathbf{P} = \mathbf{r}_{AB} \times \mathbf{P} \quad (1.4.1)$$

where $\mathbf{r}_{AB} = \mathbf{r}_B - \mathbf{r}_A$ is the relative displacement vector from A to B . Since $\mathbf{r}_{AB}$ depends strictly on the relative position of A and B , $\mathbf{G}_O$ is independent of O .

2. Moment About Any Arbitrary Base Point O' :

Let O' be any second reference point with position vector $\mathbf{r}_{O'}$ relative to O . The position vectors of A and B relative to O' are:

$$\mathbf{r}'_A = \mathbf{r}_A - \mathbf{r}_{O'}, \quad \mathbf{r}'_B = \mathbf{r}_B - \mathbf{r}_{O'}$$

Calculating the couple moment $\mathbf{G}_{O'}$ about O' :

$$\begin{aligned} \mathbf{G}_{O'} &= \mathbf{r}'_A \times (-\mathbf{P}) + \mathbf{r}'_B \times \mathbf{P} \\ &= (\mathbf{r}_A - \mathbf{r}_{O'}) \times (-\mathbf{P}) + (\mathbf{r}_B - \mathbf{r}_{O'}) \times \mathbf{P} \\ &= -\mathbf{r}_A \times \mathbf{P} + \mathbf{r}_{O'} \times \mathbf{P} + \mathbf{r}_B \times \mathbf{P} - \mathbf{r}_{O'} \times \mathbf{P} \\ &= (\mathbf{r}_B - \mathbf{r}_A) \times \mathbf{P} = \mathbf{r}_{AB} \times \mathbf{P} = \mathbf{G}_O \end{aligned} \quad (1.4.2)$$

3. Conclusion:

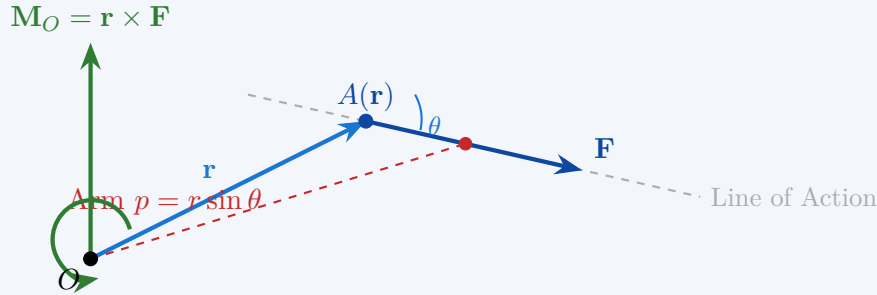
Since $\mathbf{G}_{O'} = \mathbf{G}_O$, the vector moment of a couple is completely invariant under shift of reference origin. The magnitude of the couple moment is $G = |\mathbf{G}_O| = P \cdot (r_{AB} \sin \theta) = P \cdot d$, where d is the perpendicular distance between the lines of action of $-\mathbf{P}$ and $+\mathbf{P}$.

1.5 Moments of a Force and Varignon's Theorem

Definition: Moment of Force About a Point and Line

1. **About a Point O :** $\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$, magnitude $M = F \cdot p$. 2. **About a Line OZ :** Let θ be angle between $\mathbf{F}$ and line OZ . Resolve $\mathbf{F}$ into $F \cos \theta$ parallel to OZ and $F \sin \theta$ perpendicular to OZ . Moment about OZ is:

$$M_{OZ} = (\mathbf{r} \times \mathbf{F}) \cdot \hat{k} \quad (1.5.1)$$



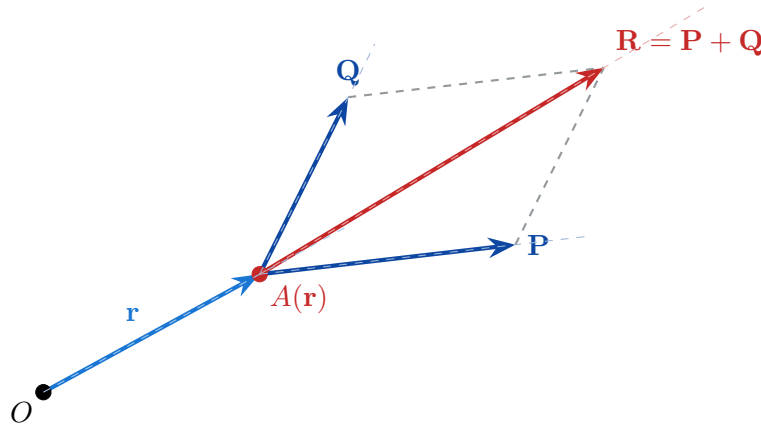
Theorem / Fundamental Law: Varignon's Theorem of Moments

The algebraic sum of the moments of any two coplanar forces about any point in their plane is equal to the moment of their resultant about that same point:

$$\mathbf{M}_O(\mathbf{R}) = \mathbf{M}_O(\mathbf{P}) + \mathbf{M}_O(\mathbf{Q}) \quad (1.5.2)$$

Rigorous Derivation / Proof: Proof of Varignon's Theorem

Let forces $\mathbf{P}$ and $\mathbf{Q}$ act at point $A(\mathbf{r})$ relative to origin O . Resultant $\mathbf{R} = \mathbf{P} + \mathbf{Q}$.



Taking vector cross product with $\mathbf{r}$:

$$\mathbf{r} \times \mathbf{R} = \mathbf{r} \times (\mathbf{P} + \mathbf{Q}) = (\mathbf{r} \times \mathbf{P}) + (\mathbf{r} \times \mathbf{Q}) \implies \mathbf{M}_O(\mathbf{R}) = \mathbf{M}_O(\mathbf{P}) + \mathbf{M}_O(\mathbf{Q})$$

1.6 Reduction of General Coplanar System of Forces

Theorem / Fundamental Law: Reduction Theorem

Any system of coplanar forces (X_i, Y_i) acting at points (x_i, y_i) on a rigid body can be reduced to a single force (X, Y) acting at origin O , together with a single couple of moment G :

$$X = \sum_{i=1}^n X_i, \quad Y = \sum_{i=1}^n Y_i, \quad G = \sum_{i=1}^n (x_i Y_i - y_i X_i) \quad (1.6.1)$$

Line of action of resultant force $R = \sqrt{X^2 + Y^2}$:

$$xY - yX = G \quad (1.6.2)$$

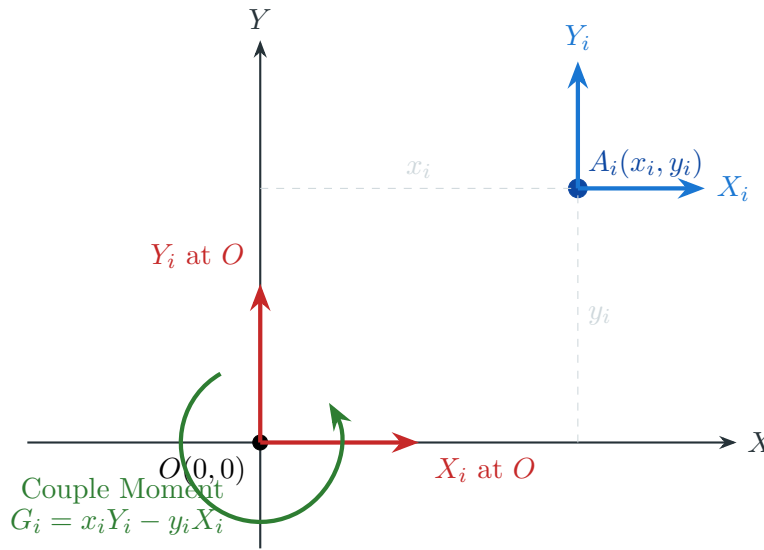


Figure 1.3: Reduction of force (X_i, Y_i) at $A_i(x_i, y_i)$ to an equivalent force (X_i, Y_i) at origin O and a couple G_i .

Rigorous Derivation / Proof: Proof of Line of Action Equation

Shift base point from $O(0,0)$ to $O'(\xi, \eta)$. New couple moment about O' becomes:

$$G' = G - \xi Y + \eta X$$

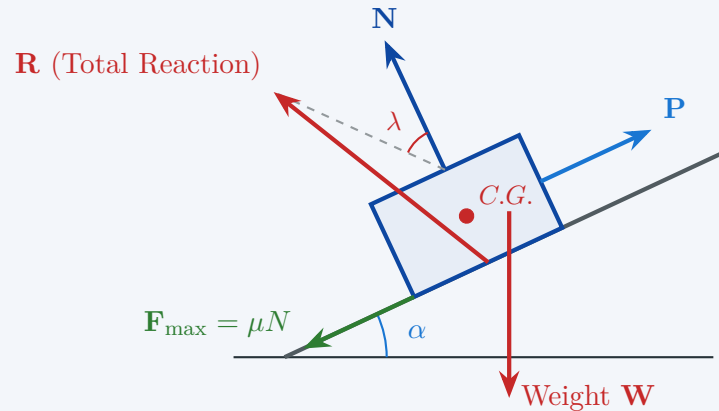
For $O'(\xi, \eta)$ to lie on line of action of resultant force R , couple moment must vanish ($G' = 0$):

$$G - \xi Y + \eta X = 0 \implies \xi Y - \eta X = G \implies xY - yX = G$$

1.7 Laws of Friction and Inclined Planes

Definition: Coulomb's Laws of Friction

Limiting friction $F_{\max} = \mu N$. Angle of friction $\lambda = \tan^{-1} \mu$. Cone of friction: semi-vertical angle λ . Minimum horizontal force to prevent sliding on rough inclined plane α : $P = W \tan(\alpha - \lambda)$.



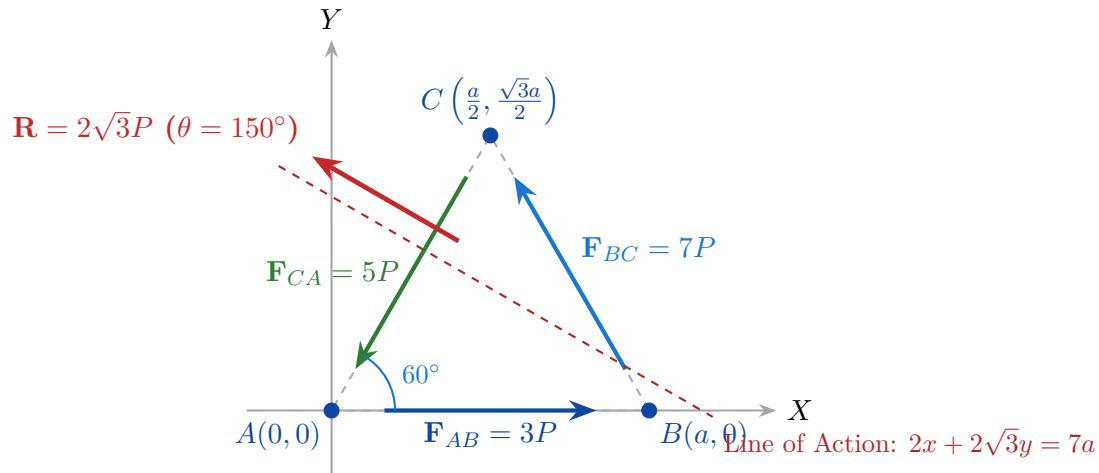
1.8 Solved Problems

Fully Solved Worked Example: Forces Along Equilateral Triangle Sides

Forces $3P, 7P, 5P$ act along sides AB, BC, CA of an equilateral triangle ABC of side a . Find resultant magnitude, angle, and line of action.

Rigorous Derivation / Proof: Solution

Origin at $A(0,0)$ with AB along X -axis.



$$X = 3P \cos 0^\circ + 7P \cos 120^\circ + 5P \cos 240^\circ = 3P + 7P \left(-\frac{1}{2}\right) + 5P \left(-\frac{1}{2}\right) = -3P$$

$$Y = 3P \sin 0^\circ + 7P \sin 120^\circ + 5P \sin 240^\circ = 0 + 7P \left(\frac{\sqrt{3}}{2}\right) + 5P \left(-\frac{\sqrt{3}}{2}\right) = \sqrt{3}P$$

Resultant magnitude:

$$R = \sqrt{X^2 + Y^2} = \sqrt{(-3P)^2 + (\sqrt{3}P)^2} = \sqrt{9P^2 + 3P^2} = 2\sqrt{3}P$$

Resultant angle:

$$\tan \theta = \frac{Y}{X} = \frac{\sqrt{3}P}{-3P} = -\frac{1}{\sqrt{3}} \implies \theta = 150^\circ$$

Moment about origin $A(0,0)$:

$$G_A = 7P \cdot (a \sin 60^\circ) = 7P \cdot \frac{\sqrt{3}a}{2} = \frac{7\sqrt{3}}{2}Pa$$

Equation of line of action ($xY - yX = G_A$):

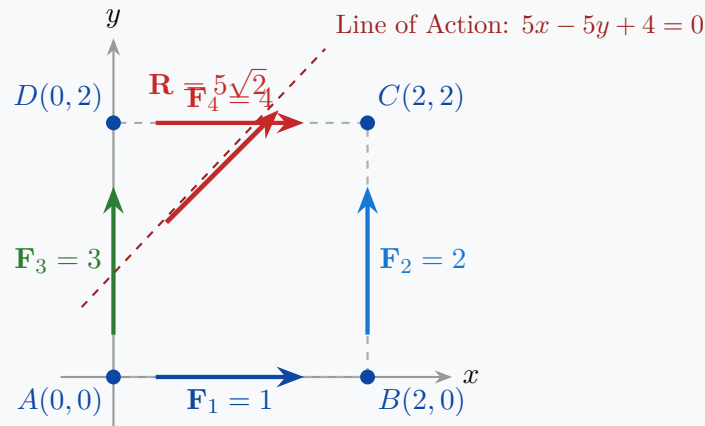
$$x(\sqrt{3}P) - y(-3P) = \frac{7\sqrt{3}}{2}Pa \implies 2x + 2\sqrt{3}y = 7a$$

Fully Solved Worked Example: Forces 1, 2, 3, 4 Along Square Sides

Forces proportional to 1, 2, 3, and 4 act along sides AB , BC , AD , and DC respectively of a square $ABCD$ of side 2 ft. Find the magnitude and line of action of their resultant.

Solution:

Take AB as x -axis, AD as y -axis with A at origin $(0,0)$. Coordinates: $A(0,0)$, $B(2,0)$, $C(2,2)$, $D(0,2)$.



Resolving components:

$$X = 1 + 4 = 5, \quad Y = 2 + 3 = 5, \quad R = \sqrt{5^2 + 5^2} = 5\sqrt{2} \text{ units}$$

Moment about $A(0,0)$: $G = 2(2) - 4(2) = -4$. Line of action $G - xY + yX = 0 \implies -4 - 5x + 5y = 0 \implies 5x - 5y + 4 = 0$.

Chapter 2

Principle of Virtual Work (Statics)

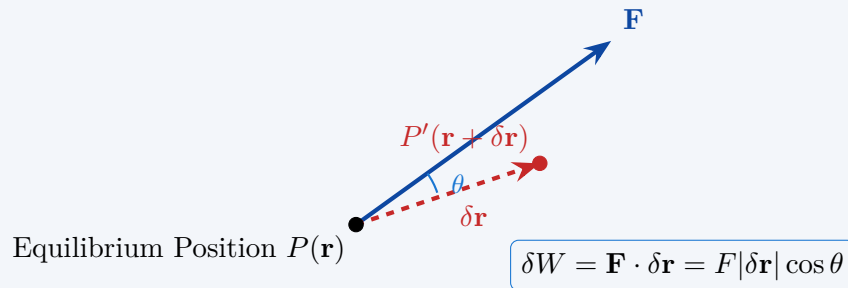
2.1 Concept of Virtual Displacement and Virtual Work

Definition: Virtual Displacement and Virtual Work

Consider a system of material points in equilibrium under a set of forces.

- A **virtual displacement** is an arbitrary, infinitesimal displacement of the system from its equilibrium position that is mathematically conceivable and consistent with the geometrical constraints imposed on the system, taking place in zero time.
- **Virtual Work (δW)**: The work done by a force $\mathbf{F}$ during a virtual displacement $\delta \mathbf{r}$ is defined as the scalar product:

$$\delta W = \mathbf{F} \cdot \delta \mathbf{r} = F_x \delta x + F_y \delta y + F_z \delta z \quad (2.1.1)$$



2.2 Principle of Virtual Work

Theorem / Fundamental Law: Principle of Virtual Work for Coplanar Systems

A system of rigid bodies connected together and acted upon by coplanar forces is in **equilibrium** if and only if the total virtual work done by all the active forces vanishes for every small arbitrary virtual displacement of the system consistent with its geometrical constraints:

$$\sum \delta W = \sum_{i=1}^n (X_i \delta x_i + Y_i \delta y_i) = 0 \quad (2.2.1)$$

2.2.1 Active Forces vs. Forces Which Do No Work

In applying the Principle of Virtual Work, internal constraint forces which perform zero net virtual work can be omitted:

Forces which Do NO Virtual Work ($\delta W = 0$)

1. Reaction of a smooth fixed surface or support (perpendicular to displacement).
2. Reaction at a smooth fixed pin, hinge, or fulcrum (displacement of point of application is zero).
3. Tension of an inextensible light string or thrust of a rigid light rod connecting two fixed points ($\delta l = 0$).
4. Normal reaction between two smooth bodies in contact.
5. Force of friction when one body rolls without slipping on another surface.

Active Forces Which DO Virtual Work

1. **Weight of a Body (W):** $\delta W = -W \delta y_{cg}$ (where y_{cg} is measured vertically upward).
2. **Tension in an Elastic String (T):** $\delta W = -T \delta l$ (where l is the stretched length).
3. **Tension in an Inextensible String (T):** $\delta W = -T \delta s$ (when length s is varied).
4. **Thrust in a Rod (S):** $\delta W = +S \delta x$ (where x is the length of the rod).

2.3 Hooke's Law for Elastic Strings

Definition: Hooke's Law

The tension T in an elastic string or spring of natural length a stretched to a length l is proportional to its extension ($l - a$):

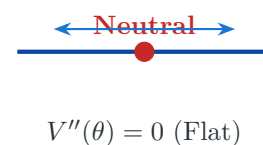
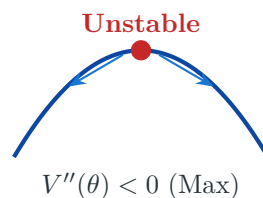
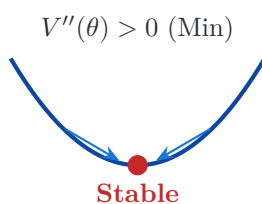
$$T = \lambda \left(\frac{l - a}{a} \right) \quad (2.3.1)$$

where λ is the **modulus of elasticity**.

2.4 Stability of Equilibrium

Let the potential energy of a conservative system in terms of a single generalized coordinate θ be $V(\theta)$.

- **Equilibrium Condition:** $\frac{dV}{d\theta} = 0$.
- **Stable Equilibrium:** $\frac{d^2V}{d\theta^2} > 0$ (Potential energy V is a minimum).
- **Unstable Equilibrium:** $\frac{d^2V}{d\theta^2} < 0$ (Potential energy V is a maximum).
- **Neutral Equilibrium:** $\frac{d^2V}{d\theta^2} = 0$.

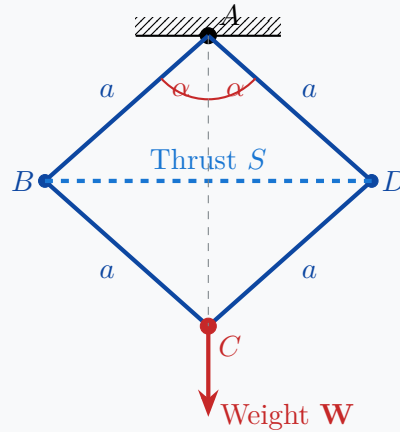


2.5 Worked Examples (Complete Unabridged Collection)

Fully Solved Worked Example: Rhombus of Four Rods with Central Thrust

Four equal uniform light rods each of length a are freely jointed to form a rhombus $ABCD$. The system is suspended from joint A , and a light rod BD holds the frame in shape with $\angle BAD = 2\alpha$. A weight W is attached to the lowest joint C . Show that the thrust S in the rod BD is $2W \tan \alpha$.

Solution:



1. Express Coordinates in Terms of Angle α :

Let α be half the vertical angle at joint A . The depth of joint C below A is $y_C = 2a \cos \alpha$. The length of horizontal diagonal BD is $x_{BD} = 2a \sin \alpha$.

Set Up Virtual Work Equation: For a small virtual variation $\delta\alpha$, the active forces doing work are weight W at C and thrust S in rod BD :

$$\delta W_{\text{net}} = W \delta y_C + S \delta x_{BD} = 0$$

Compute Variations:

$$\delta y_C = \delta(2a \cos \alpha) = -2a \sin \alpha \delta\alpha$$

$$\delta x_{BD} = \delta(2a \sin \alpha) = 2a \cos \alpha \delta\alpha$$

Substitute and Solve for Thrust S :

$$W(-2a \sin \alpha \delta\alpha) + S(2a \cos \alpha \delta\alpha) = 0$$

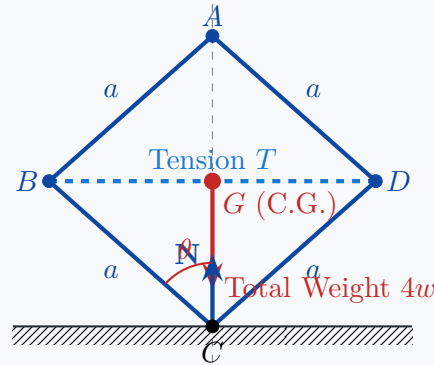
$$-2Wa \sin \alpha + 2Sa \cos \alpha = 0 \implies S = W \frac{\sin \alpha}{\cos \alpha} = W \tan \alpha$$

(Note: If four heavy rods each of weight w are used, total weight at center is $4w + W$, giving $S = (2w + W) \tan \alpha$.)

Fully Solved Worked Example: Rhombus Resting on Horizontal Plane

Four equal uniform heavy rods each of weight w are jointed to form a rhombus $ABCD$. The rhombus rests in a vertical plane with C on a smooth horizontal table and A vertically above C . Joints B and D are connected by a string. Show that tension T in string BD is $2w \tan \theta$.

Solution:



Locate Center of Gravity: Total weight of the 4 rods is $4w$, acting at center of gravity G , which is the midpoint of AC . Height of C.G. above table C : $y_G = a \cos \theta$. Length of horizontal string BD : $l_{BD} = 2a \sin \theta$.

Write Virtual Work Equation: Giving a virtual variation $\delta\theta$:

$$-4w \delta y_G - T \delta l_{BD} = 0$$

Differentiate and Simplify:

$$\begin{aligned} -4w \delta(a \cos \theta) - T \delta(2a \sin \theta) &= 0 \\ 4wa \sin \theta \delta\theta - 2Ta \cos \theta \delta\theta &= 0 \implies T = 2w \tan \theta \end{aligned}$$

Fully Solved Worked Example: Parallelogram of Four Rods Suspended by Joint

Four uniform rods are freely jointed at their extremities forming a parallelogram $ABCD$ which is suspended by joint A and kept in shape by a string AC . Prove that the tension of the string is equal to half the total weight of the four rods.

Solution:

Analyze System Geometry: Let the total weight of the four rods be W . The center of gravity G of the system is the midpoint of the diagonal string AC . Let length $AC = y$. The depth of C.G. below fixed support A is $y_G = \frac{1}{2}y$.

Apply Virtual Work: Varying length y by δy :

$$\begin{aligned} W \delta y_G - T \delta y &= 0 \\ W \delta \left(\frac{1}{2}y \right) - T \delta y &= 0 \implies \frac{1}{2}W \delta y - T \delta y = 0 \implies T = \frac{W}{2} \end{aligned}$$

Fully Solved Worked Example: Hexagon Suspended by Top Rod AB

Six equal rods AB, BC, CD, DE, EF, FA each of weight W are freely jointed forming a regular hexagon. Top rod AB is fixed horizontally, and midpoints of AB and DE are connected by a string. Prove that the tension of the string is $3W$.

Solution:

Geometric Parameters: Let side of hexagon be a . Let θ be the inclination of upper rods FA, BC to horizontal. Depth of center of gravity G of 6 rods below fixed rod AB : $y_G = \sqrt{3}a \sin \theta$. Length of string joining midpoints of AB and DE : $l = 2a \sin \theta$.

Virtual Work Equation: Total weight $6W$ acts at G :

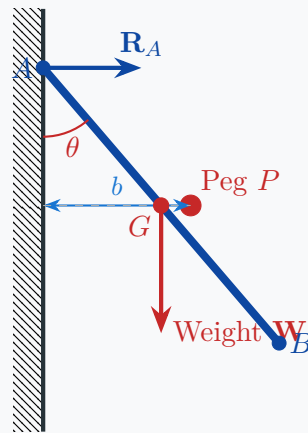
$$\begin{aligned} 6W \delta y_G - T \delta l &= 0 \\ 6W \delta(\sqrt{3}a \sin \theta) - T \delta(2a \sin \theta) &= 0 \\ 6W \sqrt{3}a \cos \theta \delta \theta - 2Ta \cos \theta \delta \theta &= 0 \implies T = 3W \end{aligned}$$

Fully Solved Worked Example: Rod Resting Against Smooth Wall and Peg

A uniform rod of length $2a$ rests in equilibrium against a smooth vertical wall and upon a smooth peg at a distance b from the wall. Show by virtual work that inclination θ of the rod to the vertical wall is given by:

$$\theta = \sin^{-1} \left(\frac{b}{a} \right)^{1/3}$$

Solution:



Express Height of Center of Gravity: Let A be the end of rod contacting wall, P be the peg, and G be the midpoint of rod. Distance along rod from wall to peg: $AP = b \csc \theta$. Distance of G from wall along rod is a . Height y_G of G above peg P :

$$y_G = a \cos \theta - b \cot \theta$$

Apply Virtual Work: For equilibrium under weight W :

$$\begin{aligned} -W \frac{dy_G}{d\theta} \delta \theta &= 0 \implies \frac{dy_G}{d\theta} = 0 \\ -a \sin \theta + b \csc^2 \theta &= 0 \implies \sin^3 \theta = \frac{b}{a} \implies \theta = \sin^{-1} \left(\frac{b}{a} \right)^{1/3} \end{aligned}$$

Fully Solved Worked Example: Elastic String on Smooth Cone

A heavy elastic string of natural length $2\pi a$ is placed horizontally around a smooth right circular cone of semi-vertical angle α with axis vertical. Show that in the position of equilibrium, the radius r of the circle formed by the string is:

$$r = a \left(1 + \frac{W \cot \alpha}{2\pi\lambda} \right)$$

where W is the weight of the string and λ is its modulus of elasticity.

Solution:

Hooke's Law for Tension: Extended perimeter of string is $2\pi r$. Stretched length extension is $2\pi r - 2\pi a$. Tension $T = \lambda \frac{2\pi r - 2\pi a}{2\pi a} = \lambda \frac{r - a}{a}$.

Depth of String below Vertex: Depth of circular line below cone vertex O : $z = r \cot \alpha \implies \delta z = \delta r \cot \alpha$.

Virtual Work Equation: Work done by gravity weight W minus work done stretching elastic string:

$$\begin{aligned} W \delta z - T \delta(2\pi r) &= 0 \\ W \cot \alpha \delta r - 2\pi T \delta r &= 0 \implies W \cot \alpha = 2\pi T \end{aligned}$$

Substitute Tension T :

$$2\pi\lambda \left(\frac{r - a}{a} \right) = W \cot \alpha \implies r - a = \frac{aW \cot \alpha}{2\pi\lambda} \implies r = a \left(1 + \frac{W \cot \alpha}{2\pi\lambda} \right)$$

Chapter 3

General Motion in Two Dimensions (Kinematics)

3.1 Cartesian Coordinate System (x, y)

Let O be a fixed origin and OX, OY be mutually perpendicular fixed axes. The position of a moving particle P at time t is specified by Cartesian coordinates (x, y) .

Rigorous Derivation / Proof: Derivation of Cartesian Velocity and Acceleration

The position vector of P relative to origin O is:

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} \quad (3.1.1)$$

where unit vectors $\mathbf{i}$ and $\mathbf{j}$ are constant in time ($\dot{\mathbf{i}} = \mathbf{0}, \dot{\mathbf{j}} = \mathbf{0}$).

1. **Velocity Vector:** Differentiating $\mathbf{r}(t)$ with respect to time t :

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} \quad (3.1.2)$$

Components of velocity parallel to OX and OY :

$$v_x = \dot{x}, \quad v_y = \dot{y} \quad (3.1.3)$$

Magnitude of velocity (speed v):

$$v = |\mathbf{v}| = \sqrt{v_x^2 + v_y^2} = \sqrt{\dot{x}^2 + \dot{y}^2} \quad (3.1.4)$$

Direction θ of velocity vector with x -axis:

$$\tan \theta = \frac{v_y}{v_x} = \frac{\dot{y}}{\dot{x}} \quad (3.1.5)$$

2. **Acceleration Vector:** Differentiating velocity $\mathbf{v}$ with respect to time t :

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2x}{dt^2}\mathbf{i} + \frac{d^2y}{dt^2}\mathbf{j} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} \quad (3.1.6)$$

Components of acceleration parallel to OX and OY :

$$a_x = \ddot{x}, \quad a_y = \ddot{y} \quad (3.1.7)$$

Magnitude of total acceleration a :

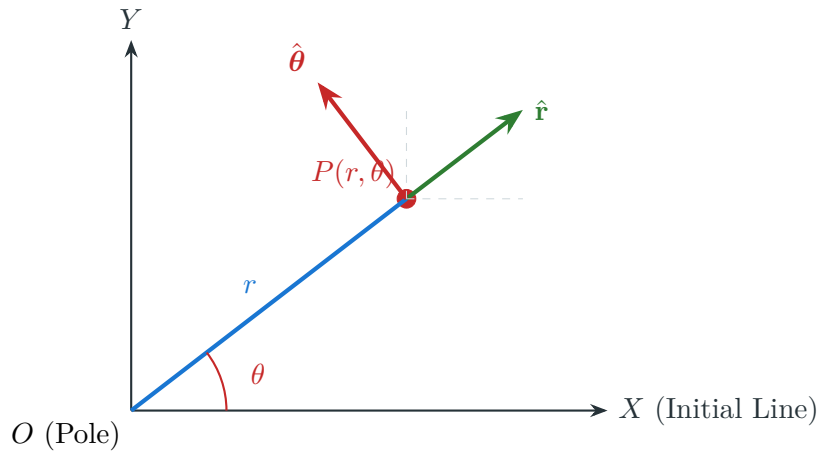
$$a = |\mathbf{a}| = \sqrt{a_x^2 + a_y^2} = \sqrt{\ddot{x}^2 + \ddot{y}^2} \quad (3.1.8)$$

3.2 Polar Coordinate System (r, θ)

Let O be the pole and OX be the initial line. The position of point P is specified by polar coordinates (r, θ) , where $r = OP$ is the radial distance and $\theta = \angle XOP$ is the vectorial angle.

Define orthogonal unit vectors:

- $\hat{\mathbf{r}}$: Unit vector along OP in direction of increasing r .
- $\hat{\boldsymbol{\theta}}$: Unit vector perpendicular to OP in direction of increasing θ .

Figure 3.1: Polar unit vectors $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ for planar motion.**Rigorous Derivation / Proof: Derivation of Unit Vector Derivatives**

Expressing unit vectors $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ in terms of Cartesian unit vectors $\mathbf{i}$ and $\mathbf{j}$:

$$\hat{\mathbf{r}} = \cos \theta \mathbf{i} + \sin \theta \mathbf{j} \quad (3.2.1)$$

$$\hat{\boldsymbol{\theta}} = -\sin \theta \mathbf{i} + \cos \theta \mathbf{j} \quad (3.2.2)$$

Differentiating equations (3.2.1) and (3.2.2) with respect to time t :

$$\frac{d\hat{\mathbf{r}}}{dt} = \frac{d}{dt}(\cos \theta \mathbf{i} + \sin \theta \mathbf{j}) = (-\sin \theta \mathbf{i} + \cos \theta \mathbf{j}) \frac{d\theta}{dt} = \dot{\theta} \hat{\boldsymbol{\theta}} \quad (3.2.3)$$

$$\frac{d\hat{\boldsymbol{\theta}}}{dt} = \frac{d}{dt}(-\sin \theta \mathbf{i} + \cos \theta \mathbf{j}) = (-\cos \theta \mathbf{i} - \sin \theta \mathbf{j}) \frac{d\theta}{dt} = -\dot{\theta} \hat{\mathbf{r}} \quad (3.2.4)$$

Rigorous Derivation / Proof: Step-by-Step Derivation of Radial and Transverse Velocity

Position vector of particle P :

$$\mathbf{r} = r \hat{\mathbf{r}}$$

Differentiating $\mathbf{r}$ with respect to time t :

$$\begin{aligned} \mathbf{v} &= \frac{d\mathbf{r}}{dt} = \frac{d}{dt}(r \hat{\mathbf{r}}) = \frac{dr}{dt} \hat{\mathbf{r}} + r \frac{d\hat{\mathbf{r}}}{dt} \\ &= \dot{r} \hat{\mathbf{r}} + r(\dot{\theta} \hat{\boldsymbol{\theta}}) \quad [\text{using eq. (3.2.3)}] \\ &= \dot{r} \hat{\mathbf{r}} + r \dot{\theta} \hat{\boldsymbol{\theta}} \end{aligned}$$

Equating components along $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$:

$$\text{Radial Velocity } v_r = \dot{r} = \frac{dr}{dt} \quad (3.2.5)$$

$$\text{Transverse Velocity } v_\theta = r \dot{\theta} = r \frac{d\theta}{dt} \quad (3.2.6)$$

$$\text{Total speed: } v = |\mathbf{v}| = \sqrt{v_r^2 + v_\theta^2} = \sqrt{\dot{r}^2 + (r\dot{\theta})^2}.$$

Rigorous Derivation / Proof: Step-by-Step Derivation of Radial and Transverse Acceleration

Differentiating velocity vector $\mathbf{v} = \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}}$ with respect to time t :

$$\begin{aligned}\mathbf{a} &= \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(\dot{r}\hat{\mathbf{r}}) + \frac{d}{dt}(r\dot{\theta}\hat{\boldsymbol{\theta}}) \\ &= \left(\ddot{r}\hat{\mathbf{r}} + \dot{r}\frac{d\hat{\mathbf{r}}}{dt}\right) + \left(\dot{r}\dot{\theta}\hat{\boldsymbol{\theta}} + r\ddot{\theta}\hat{\boldsymbol{\theta}} + r\dot{\theta}\frac{d\hat{\boldsymbol{\theta}}}{dt}\right)\end{aligned}$$

Substituting $\frac{d\hat{\mathbf{r}}}{dt} = \dot{\theta}\hat{\boldsymbol{\theta}}$ and $\frac{d\hat{\boldsymbol{\theta}}}{dt} = -\dot{\theta}\hat{\mathbf{r}}$:

$$\begin{aligned}\mathbf{a} &= (\ddot{r}\hat{\mathbf{r}} + \dot{r}\dot{\theta}\hat{\boldsymbol{\theta}}) + (\dot{r}\dot{\theta}\hat{\boldsymbol{\theta}} + r\ddot{\theta}\hat{\boldsymbol{\theta}} - r\dot{\theta}^2\hat{\mathbf{r}}) \\ &= (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\boldsymbol{\theta}}\end{aligned}$$

Equating components along $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$:

$$\text{Radial Acceleration } a_r = \ddot{r} - r\dot{\theta}^2 \quad (3.2.7)$$

$$\text{Transverse Acceleration } a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = \frac{1}{r}\frac{d}{dt}(r^2\dot{\theta}) \quad (3.2.8)$$

3.3 Intrinsic Coordinate System (s, ψ)

Let s be the arc length along a plane curve measured from a fixed point A , and let ψ be the inclination angle of the tangent to the horizontal x -axis. Define unit vectors:

- $\hat{\mathbf{t}}$: Unit tangent vector in direction of increasing arc length s .
- $\hat{\mathbf{n}}$: Unit principal normal vector directed towards center of curvature C .

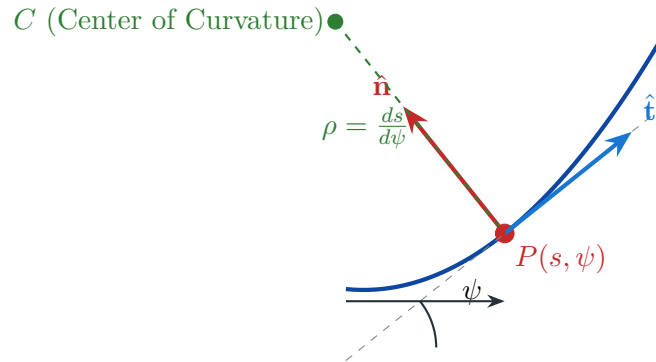


Figure 3.2: Intrinsic coordinate unit vectors $\hat{\mathbf{t}}$ and $\hat{\mathbf{n}}$, radius of curvature ρ , and center C .

Rigorous Derivation / Proof: Step-by-Step Derivation of Intrinsic Velocity

By definition of unit tangent vector:

$$\hat{\mathbf{t}} = \frac{d\mathbf{r}}{ds} \quad (3.3.1)$$

Using the chain rule for time differentiation of position vector $\mathbf{r}(t)$:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d\mathbf{r}}{ds} \frac{ds}{dt} = \left(\frac{ds}{dt} \right) \hat{\mathbf{t}} = v \hat{\mathbf{t}} \quad (3.3.2)$$

Equating components along $\hat{\mathbf{t}}$ and $\hat{\mathbf{n}}$:

$$\text{Tangential Velocity } v_t = \dot{s} = v \quad (3.3.3)$$

$$\text{Normal Velocity } v_n = 0 \quad (3.3.4)$$

This proves that the velocity vector is directed purely along the tangent to the curve.

Rigorous Derivation / Proof: Step-by-Step Derivation of Intrinsic Acceleration

Differentiating velocity vector $\mathbf{v} = v \hat{\mathbf{t}}$ with respect to time t :

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(v \hat{\mathbf{t}}) = \frac{dv}{dt} \hat{\mathbf{t}} + v \frac{d\hat{\mathbf{t}}}{dt} \quad (3.3.5)$$

To compute $\frac{d\hat{\mathbf{t}}}{dt}$, express $\hat{\mathbf{t}} = \cos \psi \mathbf{i} + \sin \psi \mathbf{j}$ and $\hat{\mathbf{n}} = -\sin \psi \mathbf{i} + \cos \psi \mathbf{j}$:

$$\frac{d\hat{\mathbf{t}}}{d\psi} = -\sin \psi \mathbf{i} + \cos \psi \mathbf{j} = \hat{\mathbf{n}}$$

Using chain rule:

$$\frac{d\hat{\mathbf{t}}}{dt} = \frac{d\hat{\mathbf{t}}}{d\psi} \frac{d\psi}{ds} \frac{ds}{dt} = \hat{\mathbf{n}} \left(\frac{1}{\rho} \right) v = \left(\frac{v}{\rho} \right) \hat{\mathbf{n}} \quad (3.3.6)$$

where $\rho = \frac{ds}{d\psi}$ is the radius of curvature.

Substituting equation (3.3.6) into equation (3.3.5):

$$\mathbf{a} = \frac{dv}{dt} \hat{\mathbf{t}} + \left(\frac{v^2}{\rho} \right) \hat{\mathbf{n}} = \ddot{s} \hat{\mathbf{t}} + \left(\frac{v^2}{\rho} \right) \hat{\mathbf{n}} \quad (3.3.7)$$

Equating components along $\hat{\mathbf{t}}$ and $\hat{\mathbf{n}}$:

$$\text{Tangential Acceleration } a_t = \frac{dv}{dt} = v \frac{dv}{ds} = \ddot{s} \quad (3.3.8)$$

$$\text{Normal Acceleration } a_n = \frac{v^2}{\rho} = v \dot{\psi} \quad (3.3.9)$$

3.4 Worked Examples (Complete Unabridged Collection)

Fully Solved Worked Example: Proportional Radial and Transverse Components

The radial and transverse velocities of a particle are proportional to each other, and its radial and transverse accelerations are also proportional to each other. Show that its velocity varies as r^k for some constant k .

Solution:

1. Use Velocity Proportionality:

Given $v_r \propto v_\theta \implies \dot{r} = c_1 r \dot{\theta}$. Dividing by r :

$$\frac{dr}{r} = c_1 d\theta \implies \ln r = c_1 \theta + c_2 \implies r = A e^{c_1 \theta}$$

2. Use Acceleration Proportionality:

Given $a_r \propto a_\theta \implies \ddot{r} - r\dot{\theta}^2 = c_2 \frac{1}{r} \frac{d}{dt}(r^2 \dot{\theta})$. Since $\dot{r} = c_1 r \dot{\theta}$, differentiating gives $\ddot{r} = c_1 \dot{r} \dot{\theta} + c_1 r \ddot{\theta} = c_1^2 r \dot{\theta}^2 + c_1 r \ddot{\theta}$.

3. Solve for Speed v :

Substituting into acceleration relation yields $\dot{\theta} \propto r^{k-1}$. Therefore, speed $v = \sqrt{\dot{r}^2 + r^2 \dot{\theta}^2} = \sqrt{1 + c_1^2} r \dot{\theta} \propto r^k$.

Fully Solved Worked Example: Path of Point with Two Constant Velocities is a Conic

Prove that the path of a point which possesses two constant velocities, one along a fixed direction and the other perpendicular to the radius vector drawn from a fixed point, is a conic.

Solution:

1. Resolve Velocity Components:

Let constant velocity along fixed initial line be u and perpendicular to radius vector be v . Resolving along and perpendicular to radius vector:

$$\begin{aligned}\dot{r} &= u \cos \theta \\ r\dot{\theta} &= v - u \sin \theta\end{aligned}$$

2. Form Differential Equation of Path:

Dividing the two equations:

$$\frac{dr}{r d\theta} = \frac{u \cos \theta}{v - u \sin \theta} \implies \frac{dr}{r} = \frac{u \cos \theta d\theta}{v - u \sin \theta}$$

3. Integrate to Find Orbit:

$$\ln r = -\ln(v - u \sin \theta) + \ln c \implies r(v - u \sin \theta) = c$$

Rearranging into standard polar conic form:

$$\frac{(c/v)}{r} = 1 - \left(\frac{u}{v}\right) \sin \theta$$

This represents a conic section of semi-latus rectum $l = c/v$ and eccentricity $e = u/v$.

Fully Solved Worked Example: Particle on Circle $r = 2a \cos \theta$

A particle moves along circle $r = 2a \cos \theta$ in such a way that its acceleration towards origin is always zero. Prove that $\ddot{\theta} = -2\dot{\theta}^2 \cot \theta$.

Solution:**1. Compute Radial Derivatives:**

Given $r = 2a \cos \theta$. Differentiating twice with respect to t :

$$\dot{r} = -2a \sin \theta \dot{\theta}$$

$$\ddot{r} = -2a \sin \theta \ddot{\theta} - 2a \cos \theta \dot{\theta}^2$$

2. Set Radial Acceleration to Zero:

$$a_r = \ddot{r} - r\dot{\theta}^2 = 0$$

$$-2a \sin \theta \ddot{\theta} - 2a \cos \theta \dot{\theta}^2 - (2a \cos \theta) \dot{\theta}^2 = 0$$

$$-2a \sin \theta \ddot{\theta} - 4a \cos \theta \dot{\theta}^2 = 0$$

3. Solve for $\ddot{\theta}$:

$$2a \sin \theta \ddot{\theta} = -4a \cos \theta \dot{\theta}^2 \implies \ddot{\theta} = -2\dot{\theta}^2 \cot \theta$$

Fully Solved Worked Example: Uniform Speed on Catenary Curve

A particle describes a curve with uniform speed v . If the magnitude of its total acceleration at any arc length s is given by $a = \frac{v^2 c}{s^2 + c^2}$, prove that the path described by the particle is an intrinsic catenary $s = c \tan \psi$.

Solution:**1. Analyze Acceleration Components:**

Since speed $v = \text{constant}$, tangential acceleration $a_t = \dot{v} = 0$. The total acceleration equals normal acceleration a_n :

$$a = a_n = \frac{v^2}{\rho}$$

2. Equate and Determine Radius of Curvature ρ :

$$\frac{v^2}{\rho} = \frac{v^2 c}{s^2 + c^2} \implies \rho = \frac{s^2 + c^2}{c}$$

3. Integrate to Obtain Intrinsic Equation:

Since $\rho = \frac{ds}{d\psi}$:

$$\frac{ds}{d\psi} = \frac{s^2 + c^2}{c} \implies d\psi = \frac{c ds}{s^2 + c^2}$$

Integrating with $\psi = 0$ at $s = 0$:

$$\psi = \int_0^s \frac{c ds}{s^2 + c^2} = \tan^{-1} \left(\frac{s}{c} \right) \implies s = c \tan \psi$$

which is the intrinsic equation of a common catenary.

Fully Solved Worked Example: Bead on Smooth Cardioid Wire

A small bead slides with constant speed v on a smooth wire in the shape of a cardioid $r = a(1 + \cos \theta)$. Show that $\dot{\theta} = \frac{v}{2a} \sec(\theta/2)$ and radial acceleration is constant.

Solution:**1. Differentiate Cardioid Equation:**

Given $r = a(1 + \cos \theta) = 2a \cos^2(\theta/2)$.

$$\dot{r} = -a \sin \theta \dot{\theta} = -2a \sin(\theta/2) \cos(\theta/2) \dot{\theta}$$

2. Relate Speed v to $\dot{\theta}$:

$$\begin{aligned} v^2 &= \dot{r}^2 + r^2 \dot{\theta}^2 = a^2 \dot{\theta}^2 \sin^2 \theta + a^2 (1 + \cos \theta)^2 \dot{\theta}^2 \\ &= a^2 \dot{\theta}^2 [\sin^2 \theta + 1 + 2 \cos \theta + \cos^2 \theta] = 2a^2 (1 + \cos \theta) \dot{\theta}^2 \\ &= 4a^2 \cos^2(\theta/2) \dot{\theta}^2 \end{aligned}$$

Taking positive square root:

$$v = 2a \cos(\theta/2) \dot{\theta} \implies \dot{\theta} = \frac{v}{2a} \sec(\theta/2)$$

3. Compute Radial Acceleration a_r :

$$a_r = \ddot{r} - r \dot{\theta}^2 = -\frac{3v^2}{4a} = \text{constant}$$

Chapter 4

Simple Harmonic Motion (Dynamics)

4.1 Introduction and Governing Differential Equation

Definition: Simple Harmonic Motion (SHM)

A particle moving along a straight line is said to execute **Simple Harmonic Motion** if its acceleration is always directed towards a fixed point on the line (called the **center of oscillation** or **mean position**) and its magnitude is directly proportional to the displacement of the particle from that fixed point.

Let O be the mean position on line $x'Ox$, and let P be the position of the particle at time t with displacement $OP = x$. The governing differential equation of motion per unit mass is:

$$\frac{d^2x}{dt^2} = -\mu x \iff \ddot{x} + \mu x = 0 \quad (4.1.1)$$

where $\mu > 0$ is the constant of proportionality representing the acceleration per unit displacement.

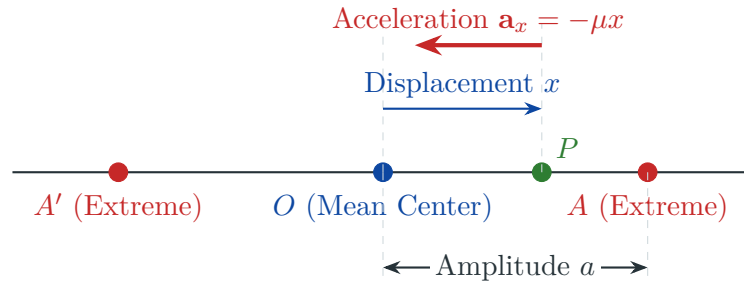


Figure 4.1: Kinematics of a particle executing Simple Harmonic Motion about mean position O .

4.2 Kinematic Expressions

4.2.1 Velocity as a Function of Position

Multiplying $\ddot{x} = -\mu x$ by $2\dot{x}$ and integrating with respect to t :

$$2\dot{x}\ddot{x} = -2\mu x\dot{x} \implies \int \frac{d}{dt}(\dot{x}^2) dt = -2\mu \int x dx \implies v^2 = -\mu x^2 + C \quad (4.2.1)$$

Let a be the maximum displacement (**amplitude**) where the particle comes to instantaneous rest ($v = 0$ at $x = a$). Then $0 = -\mu a^2 + C \implies C = \mu a^2$.

$$v^2 = \mu(a^2 - x^2) \implies v = \pm\sqrt{\mu}\sqrt{a^2 - x^2} \quad (4.2.2)$$

4.2.2 Displacement as a Function of Time

Choosing the positive sign for motion away from origin O :

$$\frac{dx}{dt} = \sqrt{\mu} \sqrt{a^2 - x^2} \implies \int \frac{dx}{\sqrt{a^2 - x^2}} = \int \sqrt{\mu} dt \quad (4.2.3)$$

- If time $t = 0$ is measured when particle is at extreme position A ($x = a$):

$$x(t) = a \cos(\sqrt{\mu}t) \quad (4.2.4)$$

- If time $t = 0$ is measured when particle passes through origin O ($x = 0$):

$$x(t) = a \sin(\sqrt{\mu}t) \quad (4.2.5)$$

4.3 Important Properties of SHM

Key Physical Quantities in SHM

1. **Amplitude** (a): Maximum displacement from the mean center O .

2. **Time Period** (T): The time taken for one complete oscillation:

$$T = \frac{2\pi}{\sqrt{\mu}} \quad (4.3.1)$$

3. **Frequency** (n): Number of complete oscillations per unit time:

$$n = \frac{1}{T} = \frac{\sqrt{\mu}}{2\pi} \quad (4.3.2)$$

4. **Maximum Speed**: $v_{\max} = a\sqrt{\mu} = \frac{2\pi a}{T}$ (achieved at mean position $x = 0$).

5. **Maximum Acceleration**: $f_{\max} = a\mu = \frac{4\pi^2 a}{T^2}$ (achieved at extreme positions $x = \pm a$).

4.4 Worked Examples (Complete Unabridged Collection)

Fully Solved Worked Example: Period from Velocities at Two Positions

A point moving in a straight line with SHM has velocities v_1 and v_2 when its distances from the center are x_1 and x_2 respectively. Show that the period of motion is:

$$T = 2\pi \sqrt{\frac{x_1^2 - x_2^2}{v_2^2 - v_1^2}}$$

Solution:

1. Apply Velocity-Displacement Relation:

For SHM, the relation between velocity v and position x is:

$$v_1^2 = \mu(a^2 - x_1^2) \quad (4.4.1)$$

$$v_2^2 = \mu(a^2 - x_2^2) \quad (4.4.2)$$

2. Eliminate Amplitude a :

Subtracting equation (4.4.1) from equation (4.4.2):

$$v_2^2 - v_1^2 = \mu(a^2 - x_2^2) - \mu(a^2 - x_1^2) = \mu(x_1^2 - x_2^2)$$

Solving for intensity constant μ :

$$\mu = \frac{v_2^2 - v_1^2}{x_1^2 - x_2^2} \Rightarrow \sqrt{\mu} = \sqrt{\frac{v_2^2 - v_1^2}{x_1^2 - x_2^2}}$$

3. Calculate Time Period T :

Using the standard period formula $T = \frac{2\pi}{\sqrt{\mu}}$:

$$T = \frac{2\pi}{\sqrt{\frac{v_2^2 - v_1^2}{x_1^2 - x_2^2}}} = 2\pi \sqrt{\frac{x_1^2 - x_2^2}{v_2^2 - v_1^2}}$$

Fully Solved Worked Example: State Parameters Calculation

A particle executing SHM has velocities u and v at two positions where its accelerations are α and β respectively. Show that the distance between these two positions is $\frac{v^2 - u^2}{\alpha + \beta}$ and the amplitude of motion is:

$$a = \sqrt{\frac{(v^2 - u^2)(\alpha^2 v^2 - \beta^2 u^2)}{(\beta^2 - \alpha^2)^2}}$$

Solution:**1. Express Accelerations and Positions:**

Let the displacements corresponding to velocities u and v be x_1 and x_2 . The magnitude of acceleration at position x is $f = \mu x$:

$$\begin{aligned}\alpha &= \mu x_1 \implies x_1 = \frac{\alpha}{\mu} \\ \beta &= \mu x_2 \implies x_2 = \frac{\beta}{\mu}\end{aligned}$$

2. Determine Distance Between Positions:

The distance between the two points is $d = x_2 - x_1$:

$$d = x_2 - x_1 = \frac{\beta - \alpha}{\mu}$$

Using $u^2 = \mu a^2 - \mu x_1^2 = \mu a^2 - \frac{\alpha^2}{\mu}$ and $v^2 = \mu a^2 - \frac{\beta^2}{\mu}$:

$$v^2 - u^2 = \frac{\alpha^2 - \beta^2}{\mu} = \frac{(\alpha - \beta)(\alpha + \beta)}{\mu}$$

Dividing by $(\alpha + \beta)$:

$$\frac{v^2 - u^2}{\alpha + \beta} = \frac{\alpha - \beta}{\mu} = x_2 - x_1 = d$$

3. Derive Amplitude a :

Substituting $\mu = \frac{\alpha^2 - \beta^2}{v^2 - u^2}$ into $v^2 = \mu a^2 - \frac{\beta^2}{\mu}$:

$$a = \sqrt{\frac{(v^2 - u^2)(\alpha^2 v^2 - \beta^2 u^2)}{(\beta^2 - \alpha^2)^2}}$$

Fully Solved Worked Example: Average Speed and Acceleration

Show that in SHM, average speed and average acceleration magnitude over a quarter period equal 0.637 times their maximum values.

Solution:

Let displacement be $x(t) = a \cos(\omega t)$ where $\omega = \sqrt{\mu}$. The velocity and acceleration magnitudes at time t are:

$$v(t) = |-a\omega \sin(\omega t)| = a\omega \sin(\omega t)$$

$$f(t) = |-a\omega^2 \cos(\omega t)| = a\omega^2 \cos(\omega t)$$

The quarter period duration is $t_q = \frac{T}{4} = \frac{\pi}{2\omega}$.

1. Average Speed over Quarter Period:

$$v_{\text{avg}} = \frac{1}{t_q} \int_0^{t_q} v(t) dt = \frac{2\omega}{\pi} \int_0^{\frac{\pi}{2\omega}} a\omega \sin(\omega t) dt$$

$$= \frac{2a\omega^2}{\pi} \left[-\frac{\cos(\omega t)}{\omega} \right]_0^{\frac{\pi}{2\omega}} = \frac{2}{\pi}(a\omega) = \frac{2}{\pi}v_{\text{max}} \approx 0.6366v_{\text{max}}$$

2. Average Acceleration Magnitude over Quarter Period:

$$f_{\text{avg}} = \frac{1}{t_q} \int_0^{t_q} f(t) dt = \frac{2\omega}{\pi} \int_0^{\frac{\pi}{2\omega}} a\omega^2 \cos(\omega t) dt$$

$$= \frac{2a\omega^3}{\pi} \left[\frac{\sin(\omega t)}{\omega} \right]_0^{\frac{\pi}{2\omega}} = \frac{2}{\pi}(a\omega^2) = \frac{2}{\pi}f_{\text{max}} \approx 0.6366f_{\text{max}}$$

Thus, $v_{\text{avg}} \approx 0.637v_{\text{max}}$ and $f_{\text{avg}} \approx 0.637f_{\text{max}}$.

Fully Solved Worked Example: SHM with Rest Points at A(a) and B(b)

A body moving in straight line OAB with SHM has zero velocity at points A and B (distances a and b from origin O). It has velocity v halfway between A and B . Show that complete period is $T = \frac{\pi(b-a)}{v}$.

Solution:**1. Identify Center of Oscillation and Amplitude:**

Since velocity is zero at $A(a)$ and $B(b)$, these points are the two extreme positions of oscillation. The center of oscillation C is the midpoint of AB :

$$x_C = \frac{a+b}{2}$$

The amplitude A_0 is half the total distance AB :

$$A_0 = \frac{b-a}{2}$$

2. Evaluate Velocity at Midpoint C:

The point halfway between A and B is the center of oscillation C , where speed reaches its maximum value:

$$v = v_{\max} = A_0\sqrt{\mu} = \left(\frac{b-a}{2}\right)\sqrt{\mu} \Rightarrow \sqrt{\mu} = \frac{2v}{b-a}$$

3. Calculate Period T :

$$T = \frac{2\pi}{\sqrt{\mu}} = \frac{2\pi}{\left(\frac{2v}{b-a}\right)} = \frac{\pi(b-a)}{v}$$

Fully Solved Worked Example: Return Time to a Point

A particle performs SHM of period T about center O . It passes through a point P ($OP = b$) with velocity v in the direction of OP . Prove that the time which elapses before it returns to P is:

$$t = \frac{T}{\pi} \tan^{-1} \left(\frac{vT}{2\pi b} \right)$$

Solution:**1. Relation Between Amplitude and Velocity at P :**

Let amplitude be a . Using $v^2 = \mu(a^2 - b^2)$:

$$a^2 - b^2 = \frac{v^2}{\mu} = \frac{v^2}{(2\pi/T)^2} = \left(\frac{vT}{2\pi} \right)^2 \implies \sqrt{a^2 - b^2} = \frac{vT}{2\pi}$$

2. Calculate Time t_1 from $P(b)$ to Extreme Position $A(a)$:

Using displacement equation $x(t) = a \cos(\sqrt{\mu}t_1)$:

$$b = a \cos(\sqrt{\mu}t_1) \implies \cos(\sqrt{\mu}t_1) = \frac{b}{a}$$

$$\tan(\sqrt{\mu}t_1) = \frac{\sqrt{a^2 - b^2}}{b} = \frac{\left(\frac{vT}{2\pi} \right)}{b} = \frac{vT}{2\pi b}$$

Taking inverse tangent:

$$\sqrt{\mu}t_1 = \tan^{-1} \left(\frac{vT}{2\pi b} \right) \implies t_1 = \frac{1}{\sqrt{\mu}} \tan^{-1} \left(\frac{vT}{2\pi b} \right)$$

3. Total Return Time:

The total time to travel from P to extreme A and back to P is $t_{\text{return}} = 2t_1$:

$$t_{\text{return}} = 2 \left(\frac{T}{2\pi} \right) \tan^{-1} \left(\frac{vT}{2\pi b} \right) = \frac{T}{\pi} \tan^{-1} \left(\frac{vT}{2\pi b} \right)$$

Fully Solved Worked Example: Period from Three Consecutive Equal Interval Measurements

A particle moves with SHM. Its distances from center at 3 consecutive seconds are x_1, x_2, x_3 . Prove that complete period is:

$$T = \frac{2\pi}{\cos^{-1}\left(\frac{x_1+x_3}{2x_2}\right)}$$

Solution:

1. Express Displacements at 1-Second Intervals:

Let time t_0 correspond to position x_2 . Then positions at $t_0 - 1$, t_0 , and $t_0 + 1$ are:

$$x_1 = a \cos(\sqrt{\mu}t_0 - \sqrt{\mu})$$

$$x_2 = a \cos(\sqrt{\mu}t_0)$$

$$x_3 = a \cos(\sqrt{\mu}t_0 + \sqrt{\mu})$$

2. Add x_1 and x_3 :

Using trigonometric identity $\cos(A - B) + \cos(A + B) = 2 \cos A \cos B$:

$$\begin{aligned} x_1 + x_3 &= a [\cos(\sqrt{\mu}t_0 - \sqrt{\mu}) + \cos(\sqrt{\mu}t_0 + \sqrt{\mu})] \\ &= 2a \cos(\sqrt{\mu}t_0) \cos(\sqrt{\mu}) = 2x_2 \cos(\sqrt{\mu}) \end{aligned}$$

3. Solve for Period T :

$$\cos(\sqrt{\mu}) = \frac{x_1 + x_3}{2x_2} \implies \sqrt{\mu} = \cos^{-1}\left(\frac{x_1 + x_3}{2x_2}\right)$$

Since $T = \frac{2\pi}{\sqrt{\mu}}$:

$$T = \frac{2\pi}{\cos^{-1}\left(\frac{x_1+x_3}{2x_2}\right)}$$

Chapter 5

Central Orbits and Kepler's Laws (Dynamics)

5.1 Central Forces and Motion in a Plane

Definition: Central Force and Central Orbit

A force $\mathbf{F}$ is called a **central force** if its line of action always passes through a fixed point O (called the center of force) and its magnitude depends solely on the distance r from O . The trajectory described by a particle moving under the action of a central force is called a **central orbit**. A central orbit always lies entirely within a single plane passing through the center of force O .

5.2 Fundamental Equations of Motion

Let O be the pole and OX be the initial line. In polar coordinates (r, θ) , the equations of motion for a particle of unit mass acted upon by central force $F(r)$ directed towards O are:

$$\text{Radial direction: } \ddot{r} - r\dot{\theta}^2 = -F(r) \quad (5.2.1)$$

$$\text{Transverse direction: } \frac{1}{r} \frac{d}{dt}(r^2\dot{\theta}) = 0 \implies r^2\dot{\theta} = h = \text{constant} \quad (5.2.2)$$

Areal Velocity and Angular Momentum

The constant $h = r^2\dot{\theta}$ represents twice the **areal velocity** (rate at which the radius vector sweeps out area):

$$\text{Areal Velocity} = \frac{1}{2}r^2\dot{\theta} = \frac{h}{2} = \text{constant} \quad (5.2.3)$$

The angular momentum per unit mass about O is strictly conserved and equals h .

5.3 Differential Equations of Central Orbits

5.3.1 Polar Form ($u = 1/r$)

Substituting $u = \frac{1}{r}$ and using $\frac{d}{dt} = hu^2 \frac{d}{d\theta}$ into equation (5.2.1) yields the famous **polar differential equation**:

$$\frac{d^2u}{d\theta^2} + u = \frac{F}{h^2u^2} \quad (5.3.1)$$

5.3.2 Pedal Form ($p = r \sin \phi$)

Let p be the length of the perpendicular drawn from origin O to the tangent at point $P(r, \theta)$. The **pedal differential equation** is:

$$\frac{h^2}{p^3} \frac{dp}{dr} = F(r) \quad (5.3.2)$$

where speed v at any point on the orbit satisfies $v = \frac{h}{p}$ and:

$$v^2 = h^2 \left[u^2 + \left(\frac{du}{d\theta} \right)^2 \right] = \frac{h^2}{p^2} \quad (5.3.3)$$

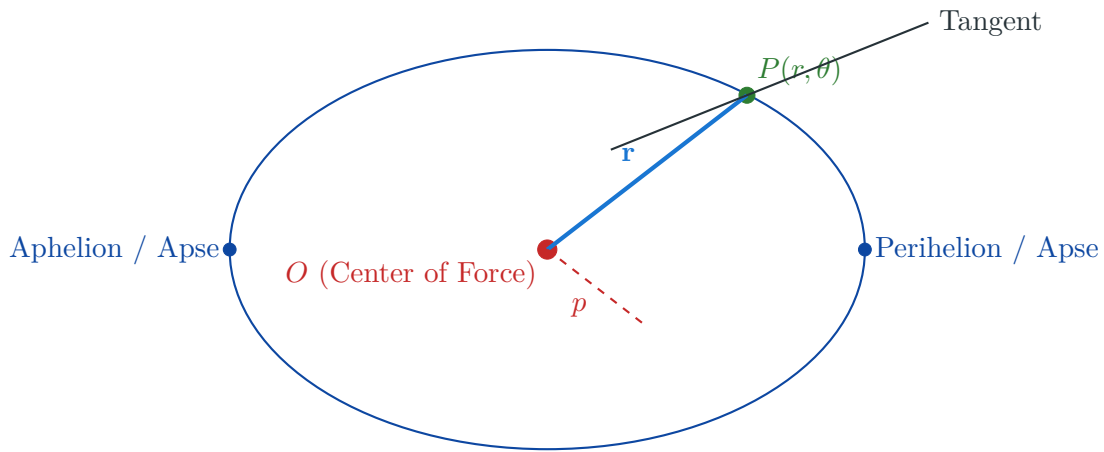


Figure 5.1: Geometry of a central orbit showing radius vector r , tangent, perpendicular distance p , and apses.

5.4 Apses and Apidal Distances

Definition: Apse

An **apse** is a point on a central orbit at which the radius vector r has a stationary value (maximum or minimum), i.e., $\frac{dr}{d\theta} = 0$.

The distance from the center of force to an apse is called an **apsidal distance**, and the angle between two consecutive apsidal vectors is the **apsidal angle**. At any apse, the tangent is perpendicular to the radius vector ($\phi = \pi/2 \implies p = r$), so speed $v = \frac{h}{r}$.

5.5 Inverse Square Law Motion ($F = \mu/r^2$)

When the central force per unit mass is attractive inverse-square $F = \frac{\mu}{r^2} = \mu u^2$:

$$\frac{d^2u}{d\theta^2} + u = \frac{\mu}{h^2}$$

The general solution is a conic section:

$$\frac{l}{r} = 1 + e \cos(\theta - \theta_0) \quad (5.5.1)$$

where semi-latus rectum $l = \frac{h^2}{\mu}$ and e is the eccentricity.

Classification of Conic Orbits

- **Elliptic Orbit** ($e < 1$): Speed equation $v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$, Periodic time $T = \frac{2\pi a^{3/2}}{\sqrt{\mu}}$, Pedal equation $\frac{b^2}{p^2} = \frac{2a}{r} - 1$.
- **Parabolic Orbit** ($e = 1$): Speed equation $v^2 = \frac{2\mu}{r}$, Pedal equation $p^2 = ar$.
- **Hyperbolic Orbit** ($e > 1$): Speed equation $v^2 = \mu \left(\frac{2}{r} + \frac{1}{a} \right)$, Pedal equation $\frac{b^2}{p^2} = \frac{2a}{r} + 1$.

5.6 Kepler's Laws of Planetary Motion**Theorem / Fundamental Law: Kepler's Three Laws**

Law I: Law of Orbits: Each planet describes an ellipse around the Sun, with the Sun situated at one of its foci.

Law II: Law of Areas: The radius vector drawn from the Sun to a planet sweeps out equal areas in equal intervals of time ($h = r^2 \dot{\theta} = \text{constant}$).

Law III: Harmonic Law: The square of the periodic time T of revolution of any planet is directly proportional to the cube of the semi-major axis a of its elliptical orbit:

$$T^2 = \left(\frac{4\pi^2}{\mu} \right) a^3 \implies T^2 \propto a^3 \quad (5.6.1)$$

5.7 Worked Examples (Complete Unabridged Collection)**Fully Solved Worked Example: Law of Force for Equiangular Spiral Orbit**

Find the law of force towards the pole for a particle describing an equiangular spiral $r = ae^{\theta \cot \alpha}$.

Solution:

Given equation of the path: $\ln r = \ln a + \theta \cot \alpha$. Differentiating with respect to θ :

$$\frac{1}{r} \frac{dr}{d\theta} = \cot \alpha \implies \tan \phi = r \frac{d\theta}{dr} = \tan \alpha \implies \phi = \alpha = \text{constant}$$

The perpendicular distance p from pole to tangent is $p = r \sin \phi = r \sin \alpha \implies \frac{1}{p^2} = \frac{1}{r^2 \sin^2 \alpha}$.
Using the pedal differential equation for central force F :

$$F = \frac{h^2}{p^3} \frac{dp}{dr} = \frac{h^2}{(r \sin \alpha)^3} (\sin \alpha) = \frac{h^2 \csc^2 \alpha}{r^3} \implies F \propto \frac{1}{r^3}$$

Fully Solved Worked Example: Law of Force for Parabolic Orbit $p^2 = ar$

A particle describes a parabolic orbit $p^2 = ar$ under a central force towards the focus. Find the law of force and speed at any point.

Solution:

Given pedal equation $p^2 = ar \implies p = a^{1/2}r^{1/2} \implies \frac{dp}{dr} = \frac{a^{1/2}}{2\sqrt{r}}$. Using pedal equation for central force F :

$$F = \frac{h^2}{p^3} \frac{dp}{dr} = \frac{h^2}{(ar)^{3/2}} \left(\frac{a^{1/2}}{2r^{1/2}} \right) = \frac{h^2}{2ar^2} \implies F = \frac{\mu}{r^2}$$

Speed $v = \frac{h}{p} \implies v^2 = \frac{h^2}{p^2} = \frac{h^2}{ar} = \frac{2\mu}{r} \implies v = \sqrt{\frac{2\mu}{r}}$.

Fully Solved Worked Example: Law of Force for Circle $r = 2a \cos \theta$

A particle describes a circle $r = 2a \cos \theta$ under a central force towards origin O . Show that central force $F \propto \frac{1}{r^5}$.

Solution:

Given $r = 2a \cos \theta \implies u = \frac{1}{2a} \sec \theta$. Differentiating: $\frac{du}{d\theta} = \frac{1}{2a} \sec \theta \tan \theta$. $\frac{d^2u}{d\theta^2} = \frac{1}{2a} (\sec^3 \theta + \sec \theta \tan^2 \theta)$. Adding u : $\frac{d^2u}{d\theta^2} + u = \frac{1}{2a} \sec \theta (\sec^2 \theta + \tan^2 \theta + 1) = \frac{1}{2a} \sec \theta (2 \sec^2 \theta) = \frac{1}{a} \sec^3 \theta = 8a^2 u^3$. Using polar differential equation $F = h^2 u^2 \left(\frac{d^2u}{d\theta^2} + u \right) = h^2 u^2 (8a^2 u^3) = 8a^2 h^2 u^5 = \frac{8a^2 h^2}{r^5}$. Thus $F \propto \frac{1}{r^5}$.

Fully Solved Worked Example: Law of Force for Lemniscate $r^2 = a^2 \cos 2\theta$

A particle describes a lemniscate $r^2 = a^2 \cos 2\theta$ under central force towards origin. Show that $F \propto \frac{1}{r^7}$.

Solution:

Pedal equation of lemniscate: $r^3 = a^2 p \implies p = \frac{r^3}{a^2} \implies \frac{dp}{dr} = \frac{3r^2}{a^2}$. Using pedal equation $F = \frac{h^2}{p^3} \frac{dp}{dr}$:

$$F = \frac{h^2}{(r^3/a^2)^3} \left(\frac{3r^2}{a^2} \right) = \frac{3a^4 h^2}{r^7} \implies F \propto \frac{1}{r^7}$$

Chapter 6

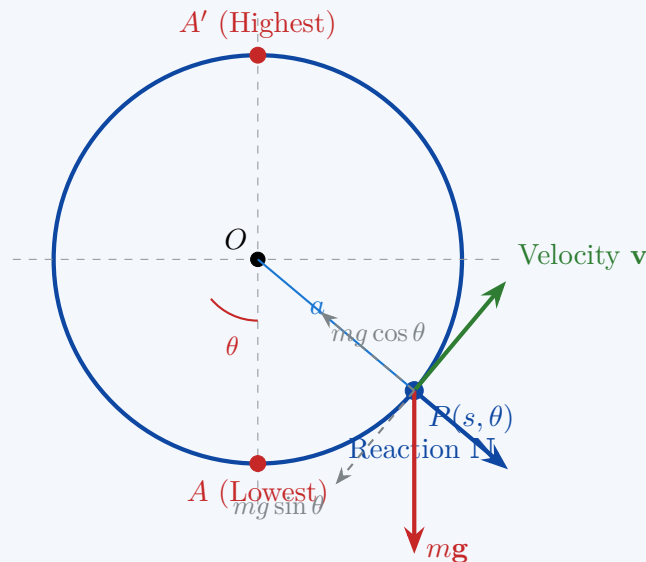
Constrained Motion on Vertical Curves

6.1 Motion on a Smooth Vertical Circle

Theorem / Fundamental Law: Vertical Circle Governing Equations

For horizontal projection velocity u from lowest point $A(\theta = 0)$ of smooth vertical circle of radius a :

1. Velocity at angle θ : $v^2 = u^2 - 2ga(1 - \cos \theta)$
2. Normal Reaction: $N = \frac{m}{a}[u^2 - 2ga + 3ga \cos \theta]$
3. Complete Loop Condition: $u \geq \sqrt{5ga}$
4. Oscillation Condition: $u \leq \sqrt{2ga}$



Rigorous Derivation / Proof: Proof of SHM About Lowest Point for Small Oscillations

For small oscillations about lowest point $A(\theta = 0)$, $\sin \theta \approx \theta$. Equation of motion along arc $s = a\theta$:

$$\ddot{s} = -g \sin \theta \approx -g \left(\frac{s}{a} \right) = - \left(\frac{g}{a} \right) s$$

This is exact **Simple Harmonic Motion** $\ddot{s} = -\mu s$ with $\mu = g/a$. Periodic time:

$$T = \frac{2\pi}{\sqrt{\mu}} = 2\pi \sqrt{\frac{a}{g}} \quad (6.1.1)$$

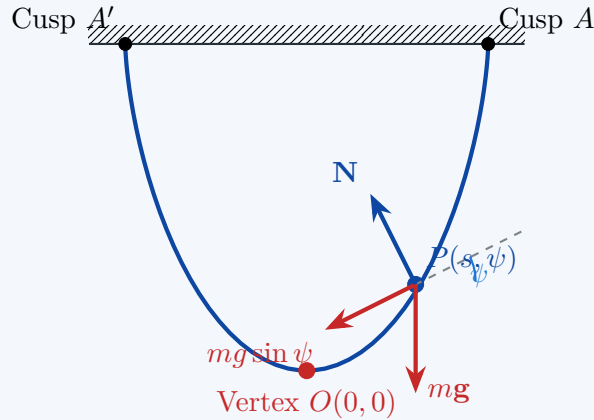
6.2 Motion on a Smooth Vertical Cycloid

Theorem / Fundamental Law: Cycloidal Pendulum

For motion along smooth cycloid $s = 4a \sin \psi$ (vertex at lowest point): Equation of motion: $\ddot{s} = -g \sin \psi = -\frac{g}{4a}s$. Exact **Simple Harmonic Motion** with periodic time:

$$T = 4\pi \sqrt{\frac{a}{g}} \quad (6.2.1)$$

The motion is **isochronous** (period is independent of amplitude).



Rigorous Derivation / Proof: Time Taken to Reach Different Heights on Cycloid

Let particle start from rest at height corresponding to arc length $s = s_0$ ($\psi = \psi_0$). Integrating equation of motion $\ddot{s} = -\omega^2 s$ (where $\omega = \frac{1}{2}\sqrt{g/a}$):

$$v^2 = \left(\frac{ds}{dt}\right)^2 = \omega^2(s_0^2 - s^2) \implies \frac{ds}{\sqrt{s_0^2 - s^2}} = -\omega dt$$

Time t taken to fall from s_0 to arc length s :

$$t = \frac{1}{\omega} \cos^{-1} \left(\frac{s}{s_0} \right) = 2\sqrt{\frac{a}{g}} \cos^{-1} \left(\frac{\sin \psi}{\sin \psi_0} \right)$$

Time taken to reach the vertex ($s = 0$): $t_{\text{vertex}} = \frac{\pi}{2\omega} = \pi\sqrt{\frac{a}{g}}$.

Chapter 7

Motion in a Moving Frame of Reference (Dynamics)

7.1 Inertial vs. Non-Inertial Frames

Definition: Frames of Reference

A **frame of reference** is a coordinate system used to specify the position and motion of physical bodies.

1. **Inertial (Newtonian) Frame:** A reference frame at rest or moving with constant linear velocity (zero acceleration) relative to fixed space. Newton's second law holds strictly in an inertial frame:

$$m\mathbf{a}_{\text{fixed}} = \mathbf{F}_{\text{external}} \quad (7.1.1)$$

2. **Non-Inertial Frame:** An accelerating or rotating reference frame. Newton's second law is valid in a non-inertial frame only if fictitious (pseudo) forces are introduced.

7.2 Motion in a Linearly Accelerating Frame

Let OXY be a fixed inertial frame and $O'X'Y'$ be a frame accelerating linearly with acceleration $\mathbf{a}_0$ relative to OXY . If $\mathbf{r}$ and $\mathbf{r}'$ are position vectors of a particle of mass m in the fixed and moving frames respectively:

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{r}' \implies \mathbf{a} = \mathbf{a}_0 + \mathbf{a}' \quad (7.2.1)$$

Equation of motion in the accelerating frame:

$$m\mathbf{a}' = \mathbf{F}_{\text{external}} - m\mathbf{a}_0 \quad (7.2.2)$$

where $-m\mathbf{a}_0$ is the **fictitious (inertial) force**.

7.3 Motion in a Uniformly Rotating Frame

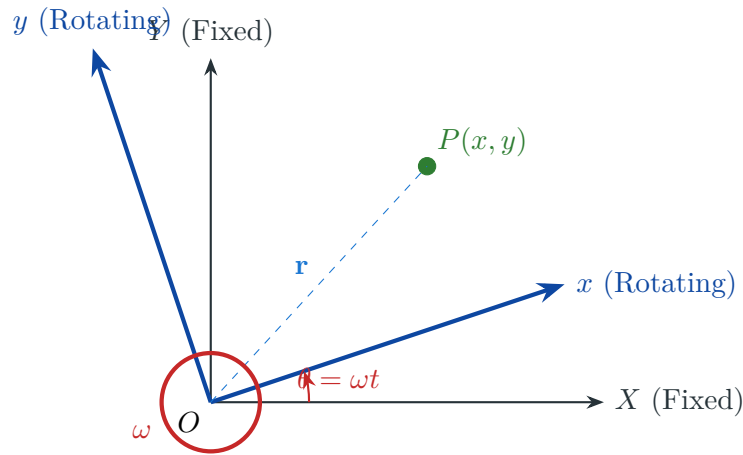
Consider a frame of reference Oxy rotating with uniform angular velocity $\omega = \omega\hat{k}$ about an axis through origin O , perpendicular to the xy -plane, relative to fixed frame OXY .

Let $\hat{i}, \hat{j}$ be unit vectors along the rotating x and y axes. The position vector is $\mathbf{r} = x\hat{i} + y\hat{j}$.

7.3.1 Transformation of Velocity

The rate of change of unit vectors in fixed space is:

$$\frac{d\hat{i}}{dt} = \omega \times \hat{i} = \omega\hat{j}, \quad \frac{d\hat{j}}{dt} = \omega \times \hat{j} = -\omega\hat{i} \quad (7.3.1)$$

Figure 7.1: Fixed coordinate frame OXY and rotating coordinate frame Oxy .

Velocity in the fixed frame:

$$\mathbf{v}_{\text{fixed}} = (\dot{x} - \omega y)\hat{i} + (\dot{y} + \omega x)\hat{j} = \mathbf{v}_{\text{rel}} + \omega \times \mathbf{r} \quad (7.3.2)$$

where $\mathbf{v}_{\text{rel}} = \dot{x}\hat{i} + \dot{y}\hat{j}$ is the velocity observed in the rotating frame.

7.3.2 Transformation of Acceleration

Differentiating velocity again in fixed space yields acceleration components along rotating axes:

$$a_{x,\text{fixed}} = \ddot{x} - 2\omega\dot{y} - \omega^2 x \quad (7.3.3)$$

$$a_{y,\text{fixed}} = \ddot{y} + 2\omega\dot{x} - \omega^2 y \quad (7.3.4)$$

In vector notation:

$$\mathbf{a}_{\text{fixed}} = \mathbf{a}_{\text{rel}} + 2\omega \times \mathbf{v}_{\text{rel}} + \omega \times (\omega \times \mathbf{r}) \quad (7.3.5)$$

7.4 Equations of Motion and Pseudo Forces

Theorem / Fundamental Law: Equations of Motion in Rotating Coordinates

The equations of motion for a particle of mass m acted upon by real external force $\mathbf{F} = X\hat{i} + Y\hat{j}$ in the rotating frame are:

$$m\ddot{x} = X + 2m\omega\dot{y} + m\omega^2 x \quad (7.4.1)$$

$$m\ddot{y} = Y - 2m\omega\dot{x} + m\omega^2 y \quad (7.4.2)$$

Coriolis and Centrifugal Forces

In a rotating frame, two apparent (pseudo) forces emerge:

1. Coriolis Force:

$$\mathbf{F}_{\text{Coriolis}} = -2m(\omega \times \mathbf{v}_{\text{rel}}) \quad (7.4.3)$$

Magnitude $2m\omega v_{\text{rel}}$, directed perpendicular to the relative velocity in the rotating frame. It vanishes when the particle is at rest relative to the rotating frame.

2. Centrifugal Force:

$$\mathbf{F}_{\text{Centrifugal}} = -m\omega \times (\omega \times \mathbf{r}) = m\omega^2 \mathbf{r} \quad (7.4.4)$$

Magnitude $m\omega^2 r$, directed radially outward from the axis of rotation.

7.5 Worked Examples

Fully Solved Worked Example: Bead Sliding on a Smooth Rotating Wire

A smooth straight wire rotates in a horizontal plane with uniform angular velocity ω about a fixed vertical axis through one end O . A bead of mass m is free to slide along the wire. If projected from O along the wire with initial velocity v_0 at $t = 0$, find its position $x(t)$ along the wire and normal reaction $R(t)$.

Solution:

Take the rotating wire as the x -axis ($\dot{y} = 0, y = 0$). No real force acts along the wire ($X = 0$). Equation of motion along the wire (equation (7.4.1)):

$$m\ddot{x} = m\omega^2 x \implies \ddot{x} - \omega^2 x = 0$$

General solution:

$$x(t) = C_1 \cosh(\omega t) + C_2 \sinh(\omega t)$$

Boundary conditions at $t = 0$: $x(0) = 0 \implies C_1 = 0$, and $\dot{x}(0) = v_0 \implies C_2 = \frac{v_0}{\omega}$. Thus, the position of the bead at time t is:

$$x(t) = \frac{v_0}{\omega} \sinh(\omega t)$$

Normal force $Y = R$ perpendicular to the wire (equation (7.4.2)):

$$m(0) = R - 2m\omega\dot{x} + 0 \implies R = 2m\omega\dot{x}$$

Since $\dot{x}(t) = v_0 \cosh(\omega t)$:

$$R(t) = 2m\omega v_0 \cosh(\omega t)$$

This gives the normal reaction exerted by the rotating wire on the bead.

Chapter 8

Forces in 3D, Wrench & Moment of Inertia

8.1 Reduction of 3D Systems of Forces

Theorem / Fundamental Law: Reduction of Forces in 3D

Any 3D system of forces $\mathbf{F}_i = (X_i, Y_i, Z_i)$ acting at points (x_i, y_i, z_i) reduces at origin O to a resultant force $\mathbf{R} = (X, Y, Z)$ and a resultant couple $\mathbf{G} = (L, M, N)$:

$$X = \sum X_i, \quad Y = \sum Y_i, \quad Z = \sum Z_i \quad (8.1.1)$$

$$L = \sum (y_i Z_i - z_i Y_i), \quad M = \sum (z_i X_i - x_i Z_i), \quad N = \sum (x_i Y_i - y_i X_i) \quad (8.1.2)$$

Analytical equilibrium conditions in 3D:

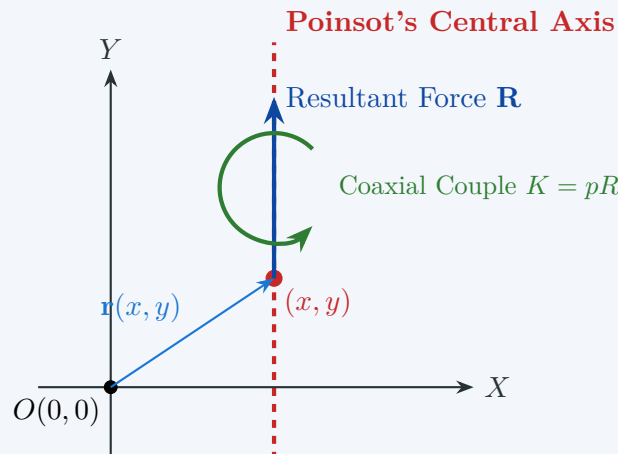
$$X = 0, \quad Y = 0, \quad Z = 0, \quad L = 0, \quad M = 0, \quad N = 0 \quad (8.1.3)$$

8.2 Dyname, Wrench and Poinso's Central Axis

Definition: Wrench and Pitch

1. **Pitch of Wrench:** $p = \frac{XL + YM + ZN}{X^2 + Y^2 + Z^2} = \frac{\mathbf{R} \cdot \mathbf{G}}{R^2}$. 2. **Wrench:** A system consisting of resultant force $\mathbf{R}$ and coaxial couple $K = pR$, denoted $[\mathbf{R}, p\mathbf{R}]$. 3. **Poinso's Central Axis Equation:**

$$\frac{L - (yZ - zY)}{X} = \frac{M - (zX - xZ)}{Y} = \frac{N - (xY - yX)}{Z} = p \quad (8.2.1)$$



Rigorous Derivation / Proof: Proof of Poinso's Central Axis Equation

Shift reference origin from $O(0,0,0)$ to a new point $O'(x,y,z)$. The new couple components (L', M', N') about O' are:

$$L' = L - (yZ - zY), \quad M' = M - (zX - xZ), \quad N' = N - (xY - yX)$$

For $O'(x,y,z)$ to lie on Poinso's Central Axis, the couple vector $\mathbf{G}' = (L', M', N')$ must be parallel to the resultant force vector $\mathbf{R} = (X, Y, Z)$:

$$\frac{L'}{X} = \frac{M'}{Y} = \frac{N'}{Z} = p$$

Substituting (L', M', N') yields the 3D Central Axis symmetric equation:

$$\frac{L - (yZ - zY)}{X} = \frac{M - (zX - xZ)}{Y} = \frac{N - (xY - yX)}{Z} = p \quad (8.2.2)$$

8.3 Moment of Inertia and Product of Inertia**Definition: Moments and Products of Inertia**

For a system of masses m_i at (x_i, y_i, z_i) :

$$I_{xx} = \sum m_i(y_i^2 + z_i^2), \quad I_{yy} = \sum m_i(x_i^2 + z_i^2), \quad I_{zz} = \sum m_i(x_i^2 + y_i^2) \quad (8.3.1)$$

$$I_{xy} = \sum m_i x_i y_i, \quad I_{yz} = \sum m_i y_i z_i, \quad I_{zx} = \sum m_i z_i x_i \quad (8.3.2)$$

Theorem / Fundamental Law: Parallel and Perpendicular Axes Theorems

1. **Parallel Axes Theorem:** $I_{L'} = I_L + Md^2$ 2. **Perpendicular Axes Theorem (Plane Lamina):** $I_z = I_x + I_y$

Rigorous Derivation / Proof: Proof of Moment of Inertia About an Inclined Line

Let OL be a line through origin O with direction cosines (α, β, γ) . The perpendicular distance p_i of mass $m_i(x_i, y_i, z_i)$ from line OL is:

$$p_i^2 = (x_i^2 + y_i^2 + z_i^2) - (\alpha x_i + \beta y_i + \gamma z_i)^2$$

Since $\alpha^2 + \beta^2 + \gamma^2 = 1$:

$$\begin{aligned} I_{OL} &= \sum m_i p_i^2 = \sum m_i [(x_i^2 + y_i^2 + z_i^2)(\alpha^2 + \beta^2 + \gamma^2) - (\alpha x_i + \beta y_i + \gamma z_i)^2] \\ &= I_{xx}\alpha^2 + I_{yy}\beta^2 + I_{zz}\gamma^2 - 2I_{xy}\alpha\beta - 2I_{yz}\beta\gamma - 2I_{zx}\gamma\alpha \end{aligned}$$

8.4 Momental Ellipsoid and Principal Axes

Rigorous Derivation / Proof: Proof of Momental Ellipsoid Equation

Let $P(x, y, z)$ be a point on line OL at distance r from origin O such that $r = 1/\sqrt{I_{OL}}$. Then $x = r\alpha, y = r\beta, z = r\gamma \implies \alpha = x/r, \beta = y/r, \gamma = z/r$. Substituting into I_{OL} :

$$I_{OL} = I_{xx} \left(\frac{x}{r}\right)^2 + I_{yy} \left(\frac{y}{r}\right)^2 + I_{zz} \left(\frac{z}{r}\right)^2 - 2I_{xy} \frac{xy}{r^2} - 2I_{yz} \frac{yz}{r^2} - 2I_{zx} \frac{zx}{r^2}$$

Multiplying by $r^2 = 1/I_{OL}$ gives the quadric equation of the **Momental Ellipsoid**:

$$I_{xx}x^2 + I_{yy}y^2 + I_{zz}z^2 - 2I_{xy}xy - 2I_{yz}yz - 2I_{zx}zx = 1 \quad (8.4.1)$$

The principal axes of this ellipsoid are the **Principal Axes of Inertia**, along which products of inertia vanish ($I_{xy} = I_{yz} = I_{zx} = 0$).

8.5 Standard Moments of Inertia Table

Theorem / Fundamental Law: Standard M.I. Formula Table

1. **Thin Rod (Length $2a$):** $I_{\text{perp}} = \frac{1}{3}ma^2$
2. **Rectangular Plate ($2a \times 2b$):** $I_x = \frac{1}{3}mb^2, I_y = \frac{1}{3}ma^2, I_z = \frac{1}{3}m(a^2 + b^2)$
3. **Circular Ring (Radius a):** $I_{\text{axis}} = ma^2, I_{\text{dia}} = \frac{1}{2}ma^2$
4. **Circular Disc (Radius a):** $I_{\text{axis}} = \frac{1}{2}ma^2, I_{\text{dia}} = \frac{1}{4}ma^2$
5. **Solid Sphere (Radius a):** $I = \frac{2}{5}ma^2$
6. **Spherical Shell (Radius a):** $I = \frac{2}{3}ma^2$
7. **Solid Cylinder (Radius a , Length $2h$):** $I_{\text{axis}} = \frac{1}{2}ma^2, I_{\text{perp}} = m\left(\frac{a^2}{4} + \frac{h^2}{3}\right)$
8. **Solid Cone (Base Radius a , Height h):** $I_{\text{axis}} = \frac{3}{10}ma^2$

Chapter 9

Master Formula Cheat Sheet

9.1 Statics, Friction & Virtual Work

Statics & Equilibrium Reference

1. Coplanar Resultant & Line of Action:

2.

$$R = \sqrt{X^2 + Y^2}, \quad \tan \theta = \frac{Y}{X}, \quad yX - xY = G$$

3. Lami's Theorem:

4.

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$$

5. Limiting Friction & Angle of Friction:

6.

$$F_{\max} = \mu N, \quad \tan \lambda = \mu, \quad P_{\min} = W \tan(\alpha - \lambda)$$

7. Principle of Virtual Work:

8.

$$\sum \delta W = 0 \quad (\text{Strings: } -T\delta l, \quad \text{Rods: } S\delta x, \quad \text{Gravity: } -W\delta y_{cg})$$

9. Equilibrium Stability:

10.

$$\frac{dV}{d\theta} = 0 \implies \text{Equilibrium}, \quad \frac{d^2V}{d\theta^2} > 0 \implies \text{Stable (Minimum } V)$$

9.2 Common Catenary & Kinematics

Catenary & 2D Kinematics Reference

6. Common Catenary Formulas:

$$7. \quad s = c \tan \psi, \quad y = c \cosh(x/c), \quad y^2 = c^2 + s^2, \quad T = wy, \quad T_0 = wc$$

8. Planar Kinematics (Polar & Intrinsic):

$$9. \quad v_r = \dot{r}, \quad v_\theta = r\dot{\theta}, \quad a_r = \ddot{r} - r\dot{\theta}^2, \quad a_\theta = \frac{1}{r} \frac{d}{dt}(r^2\dot{\theta})$$

$$10. \quad v_t = \dot{s}, \quad v_n = 0, \quad a_t = \dot{v} = v \frac{dv}{ds}, \quad a_n = \frac{v^2}{\rho} = v\dot{\psi}$$

9.3 Simple Harmonic Motion & Central Orbits

SHM, Central Orbits & Kepler Reference

8. Simple Harmonic Motion (SHM):

$$9. \quad \ddot{x} = -\mu x \implies v^2 = \mu(a^2 - x^2), \quad T = \frac{2\pi}{\sqrt{\mu}}, \quad x(t) = a \cos(\sqrt{\mu}t)$$

10. Central Orbits & Kepler's Laws:

$$11. \quad \frac{d^2u}{d\theta^2} + u = \frac{F}{h^2u^2}, \quad \frac{h^2}{p^3} \frac{dp}{dr} = F, \quad T^2 = \left(\frac{4\pi^2}{\mu} \right) a^3$$

9.4 Constrained Motion, 3D Rigid Bodies & Moving Frames

Constrained Motion, 3D Dynamics & Rotating Frames Reference

10. Constrained Vertical Motion:

11.

$$\text{Circle: } v^2 = u^2 - 2ag(1 - \cos \theta), \quad R = m \left[\frac{u^2}{a} - g(3 - 2 \cos \theta) \right], \quad u_{\text{loop}} \geq \sqrt{5ag}$$

12.

$$\text{Cycloid: } s = 4a \sin \psi \implies \ddot{s} = -\frac{g}{4a}s \implies T = 4\pi \sqrt{\frac{a}{g}}$$

13. Wrench & Poinsot's Central Axis:

14.

$$p = \frac{XL + YM + ZN}{X^2 + Y^2 + Z^2}, \quad \frac{L - (yZ - zY)}{X} = \frac{M - (zX - xZ)}{Y} = \frac{N - (xY - yX)}{Z} = p$$

15. Coriolis & Centrifugal Forces:

16.

$$\mathbf{F}_{\text{Coriolis}} = -2m(\boldsymbol{\omega} \times \mathbf{v}_{\text{rel}}), \quad \mathbf{F}_{\text{centrifugal}} = m\omega^2 \mathbf{r}$$